

Project 1 – Active Directory Administration Lab

Project Objective

This project demonstrates the deployment and administration of a Windows Server 2022 Active Directory environment. The lab covers domain controller configuration, user and group management, access control, and account recovery procedures. The objective is to develop practical skills commonly required for IT Support, Help Desk, and Junior System Administrator roles.

Business Scenario

A growing company requires a centralized identity management solution to manage employees across multiple departments. As a Junior IT Support Administrator, you are responsible for deploying Active Directory Domain Services (AD DS), creating organizational units, managing user accounts and security groups, implementing shared folder permissions, enforcing password policies, and supporting common Help Desk account recovery Phases.

Environment and Tools Used

- Windows Server 2022
- Windows 11
- Oracle VirtualBox
- Active Directory Domain Services (AD DS)
- Group Policy Management
- PowerShell

Key Tasks

- Installed and configured Active Directory Domain Services
- Promoted a server to Domain Controller
- Created Organizational Units (OUs)
- Managed users and security groups
- Configured shared folders and permissions
- Implemented Password and Account Lockout Policies
- Created and linked custom GPOs

- Performed password reset and account unlock procedures

Skills Demonstrated

- Active Directory Administration
- User and Group Management
- Group Policy Management
- Access Control
- Windows Server Administration
- IT Help Desk Operations

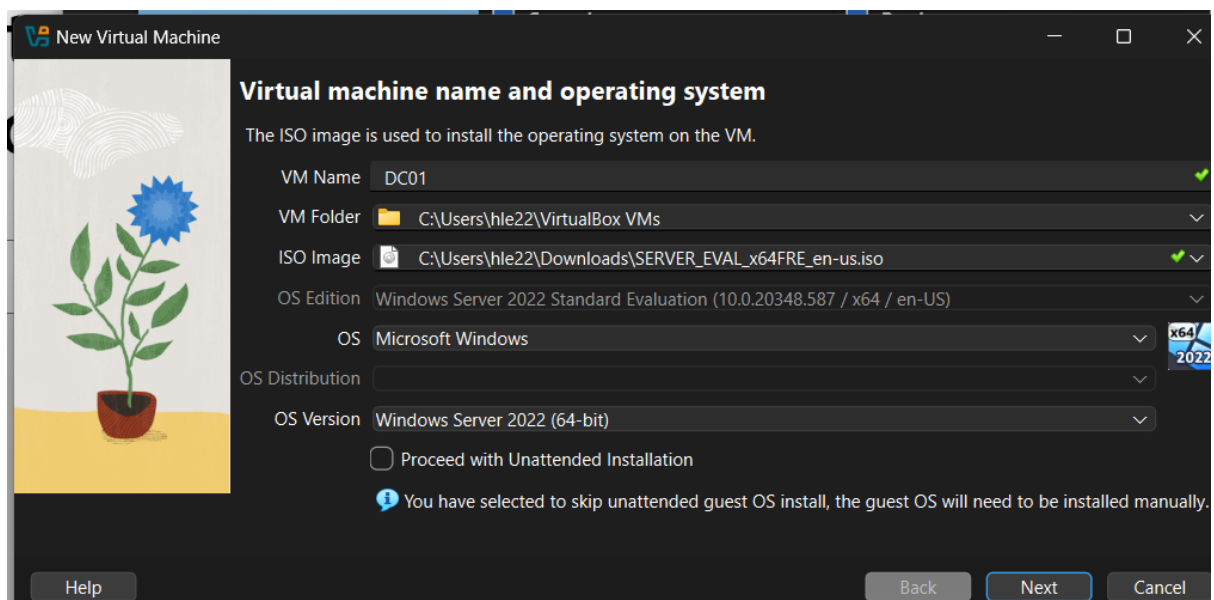
Phase 1 – Initial Windows Server Configuration

Overview

This phase focused on deploying Active Directory Domain Services (AD DS) and configuring the initial domain environment. A new domain controller was promoted to establish the corp.local domain, providing the foundation for user management, security policies, and centralized administration throughout the project.

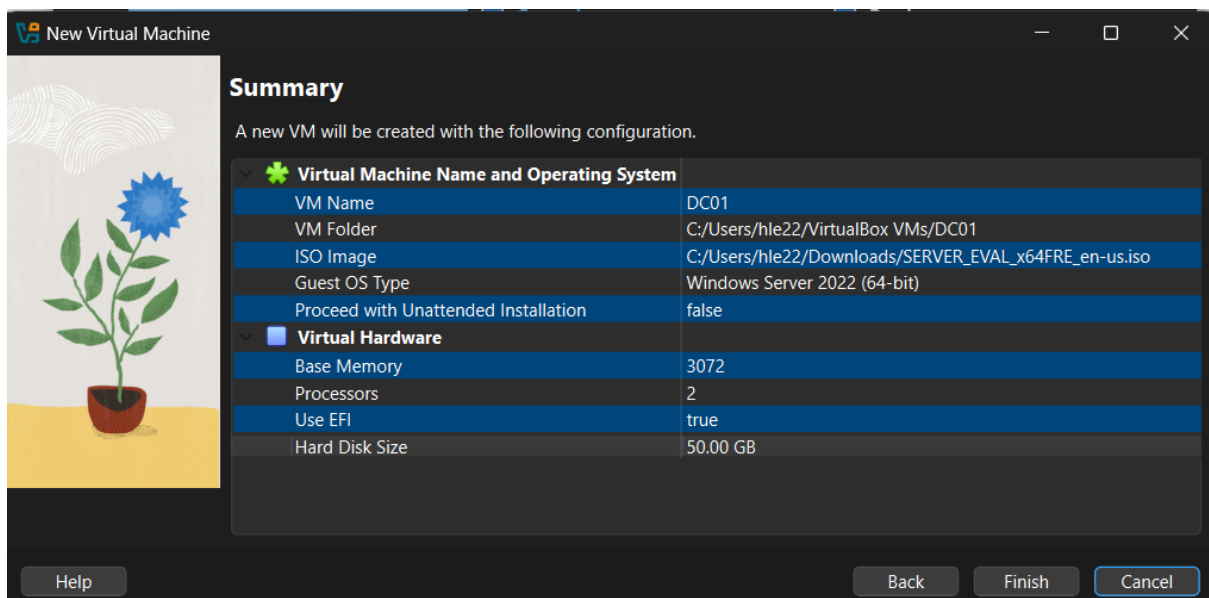
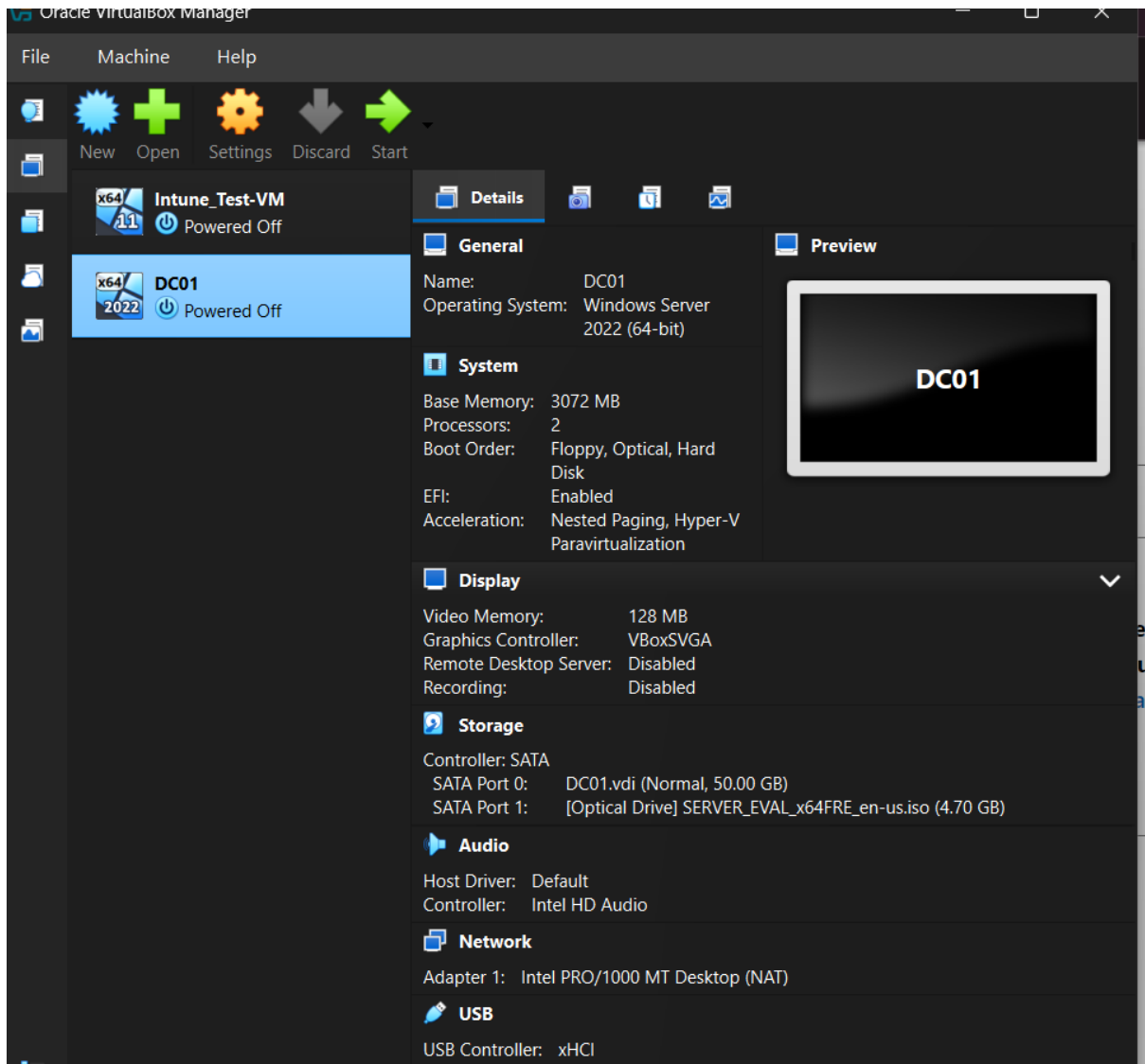
Step 1: Install Windows Server 2022

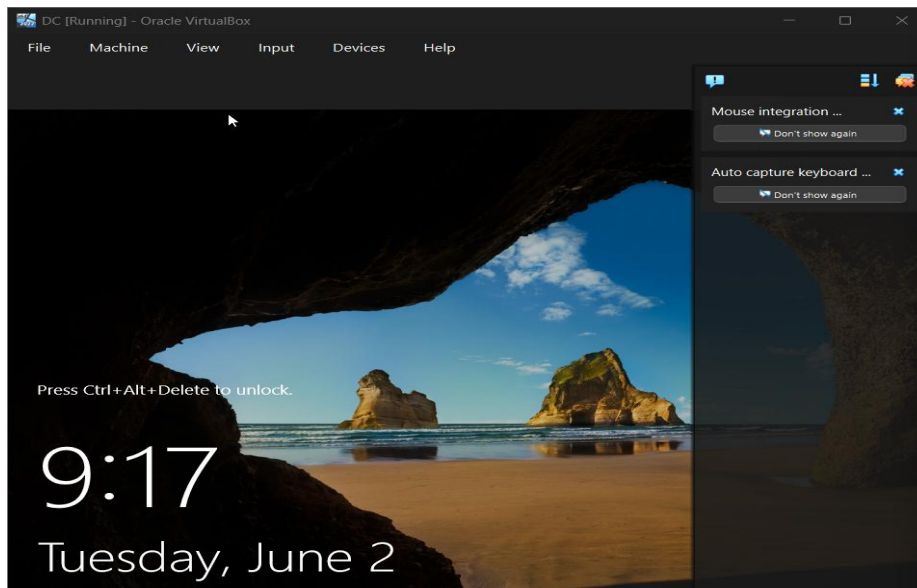
I started by creating a Windows Server 2022 virtual machine in VirtualBox. Once the operating system was installed, I logged in using the local Administrator account and confirmed that the server was ready for further configuration.



Note: The unattended installation option was disabled to allow manual configuration and provide a better understanding of the Windows Server installation process.

I then configured the virtual machine that would be used as the Domain Controller. To ensure stable performance during the lab, I allocated 3 GB of RAM, 2 virtual CPUs, and a 50 GB virtual disk before proceeding with the Windows Server installation.

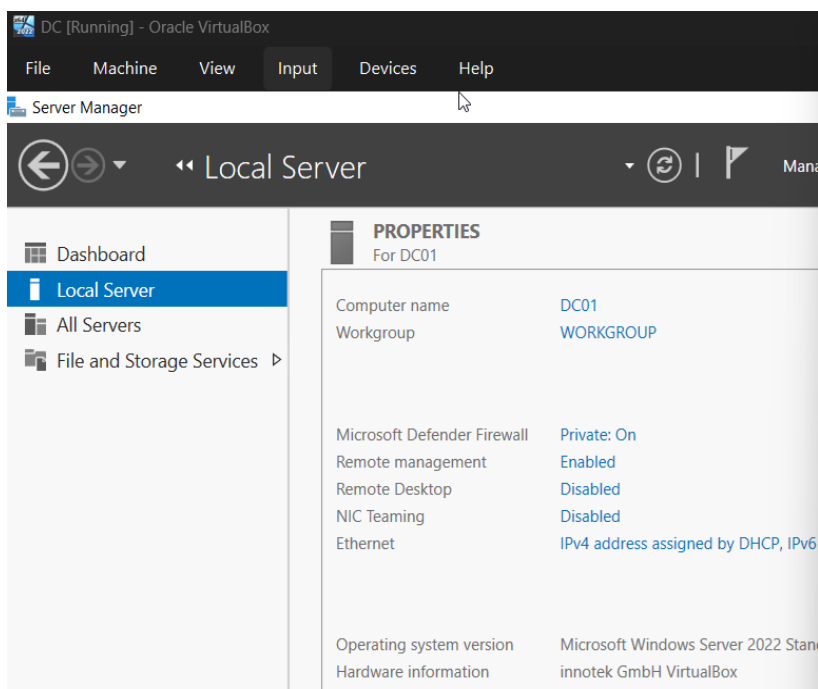




The virtual machine booted successfully into Windows Server 2022. I verified that the installation completed correctly and prepared the server for further configuration.

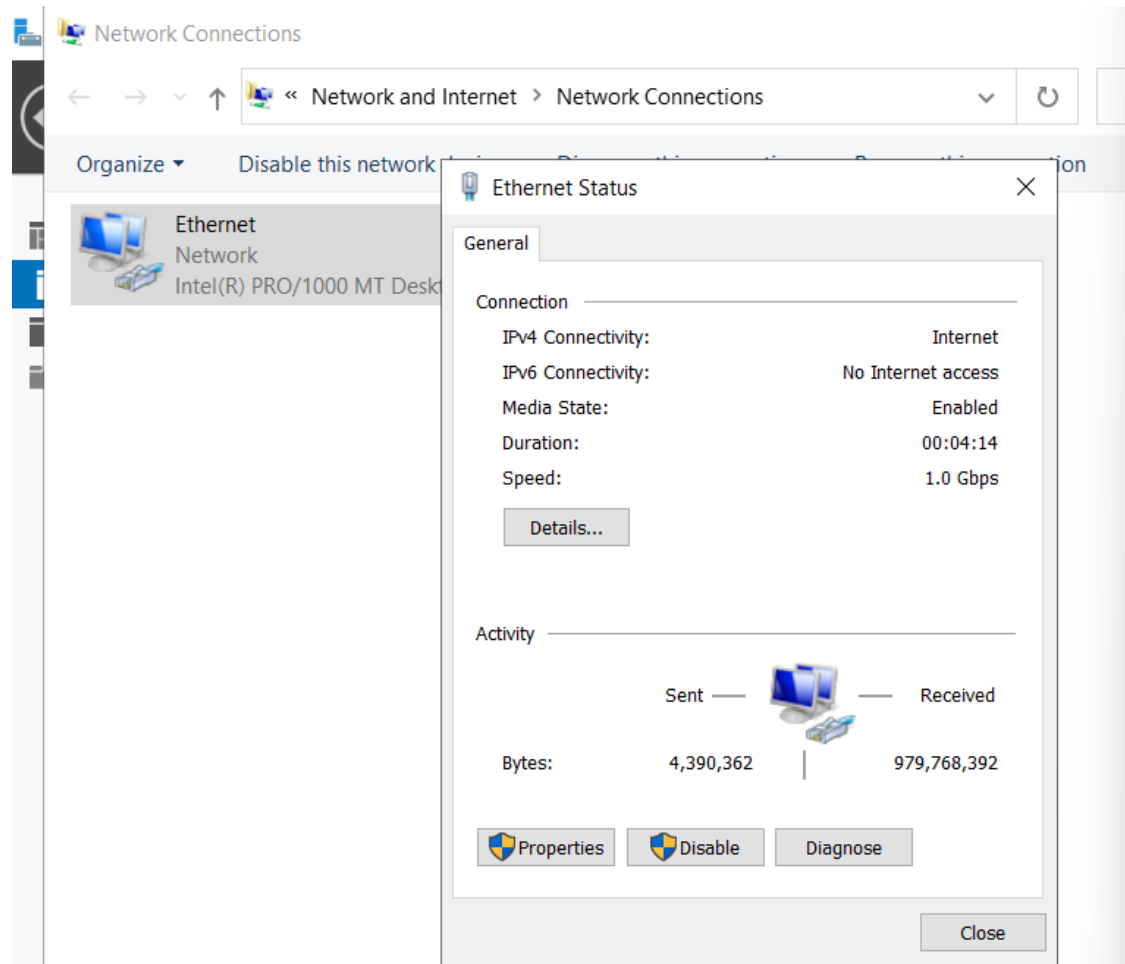
Step 2: Rename the Server

To make the server easier to identify and manage within the network, I renamed the computer from its default hostname to DC01. After restarting the system, I verified the change in Server Manager and confirmed that the new hostname was applied successfully.



Step 3: Configure Static IP Address

Before promoting the server to a Domain Controller, I reviewed the network adapter settings to prepare a static IP configuration. This ensures that clients can consistently locate the server and access Active Directory services without relying on dynamically assigned addresses.



I configured a static IP address of 10.0.2.10 with the appropriate subnet mask and default gateway. The server was also configured to use itself as the preferred DNS server, which is a common requirement for Active Directory environments.

At this stage, the server had been fully prepared for the Active Directory deployment. The hostname, IP address, and DNS settings were configured and verified, providing a stable foundation for the Domain Controller installation in the next phase.

Phase Summary

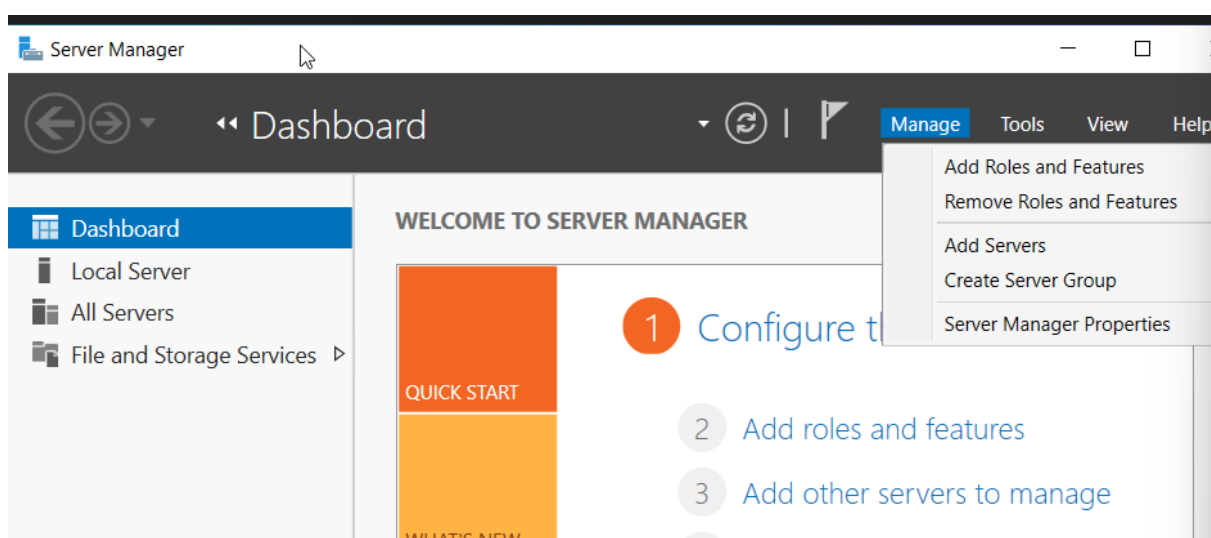
The Windows Server 2022 environment was successfully prepared for Active Directory deployment. The server was renamed to **DC01**, assigned a static IP address, and configured with the required DNS settings. These configurations provided a stable foundation for promoting the server to a Domain Controller in the next phase.

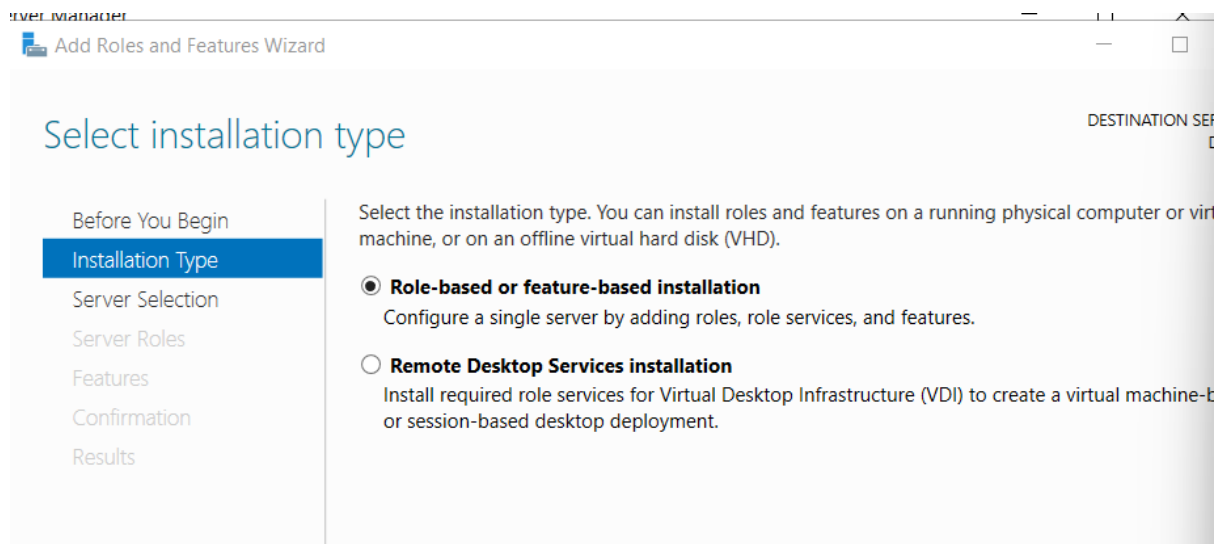
Phase 2 – Install Active Directory Domain Services (AD DS)

In this phase, I installed **Active Directory Domain Services (AD DS)** on Windows Server 2022 and promoted the server to a Domain Controller. This created the foundation of the lab environment by establishing a new Active Directory forest and enabling centralized identity management.

Step 1 – Launch Add Roles and Features Wizard

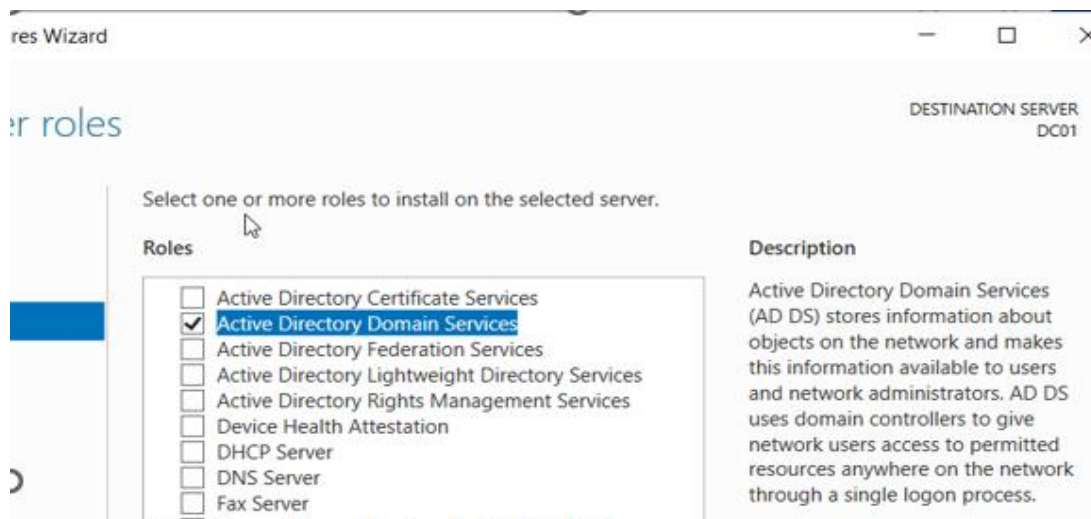
To begin the Active Directory deployment, I launched the Add Roles and Features Wizard from Server Manager. This wizard is used to install server roles and services required for domain management.



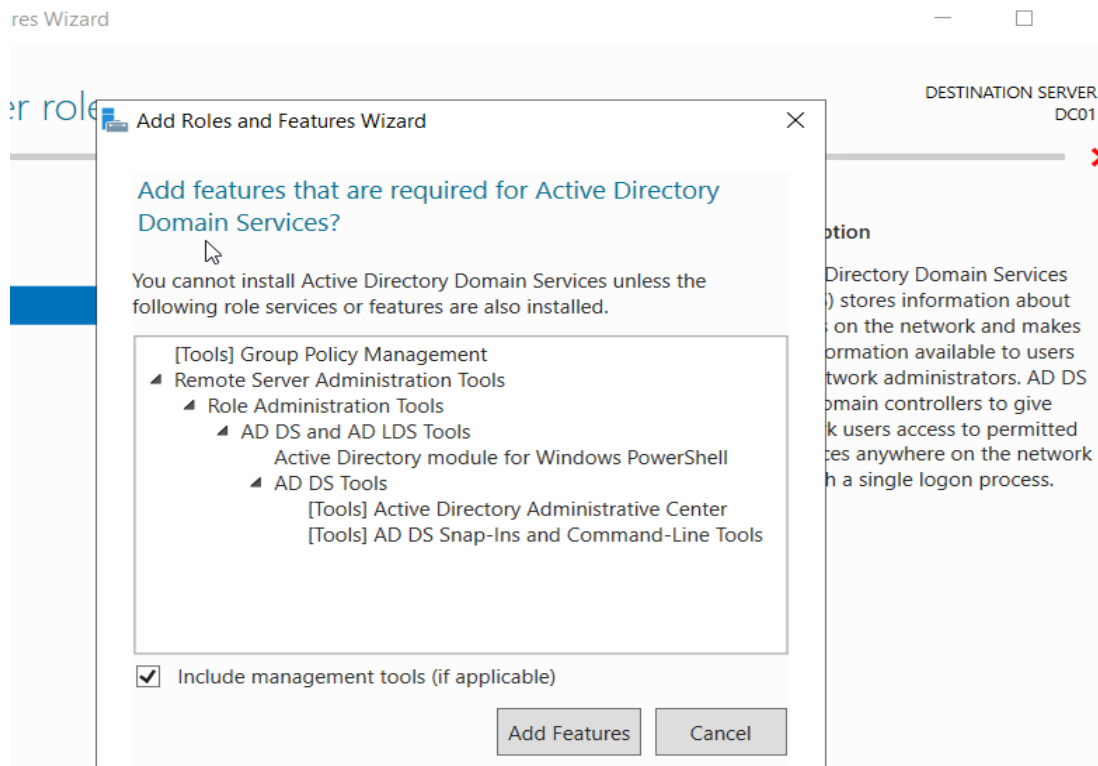


Step 2 – Install Active Directory Domain Services (AD DS)

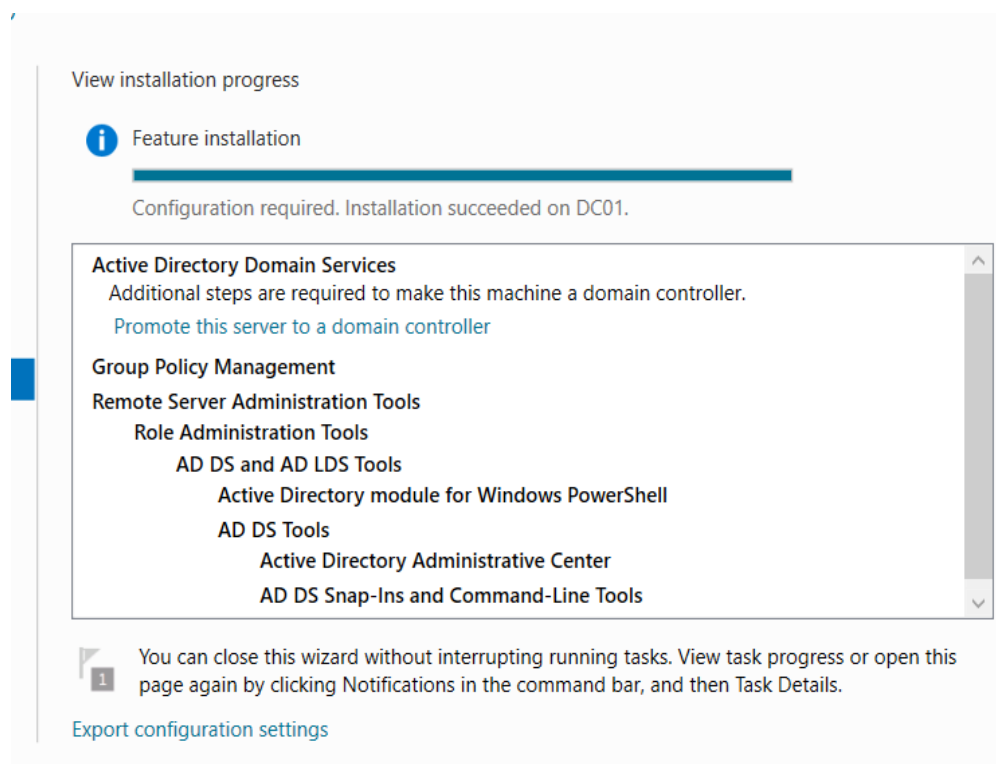
I selected the Active Directory Domain Services (AD DS) role from the list of available server roles. This role provides centralized authentication, authorization, and directory management services within the Windows Server environment.



Windows automatically prompted for additional management tools required by AD DS. I accepted the default selections to ensure the server included all administrative components needed to manage Active Directory and Group Policy.

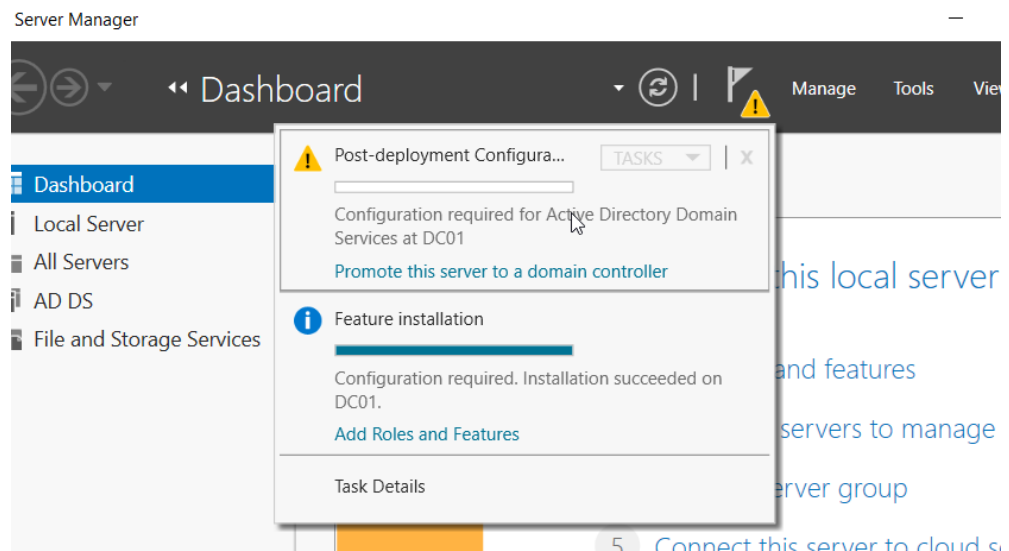


The AD DS installation completed successfully, and the server became ready for promotion to a Domain Controller. This confirmed that all required services and management tools had been installed correctly.



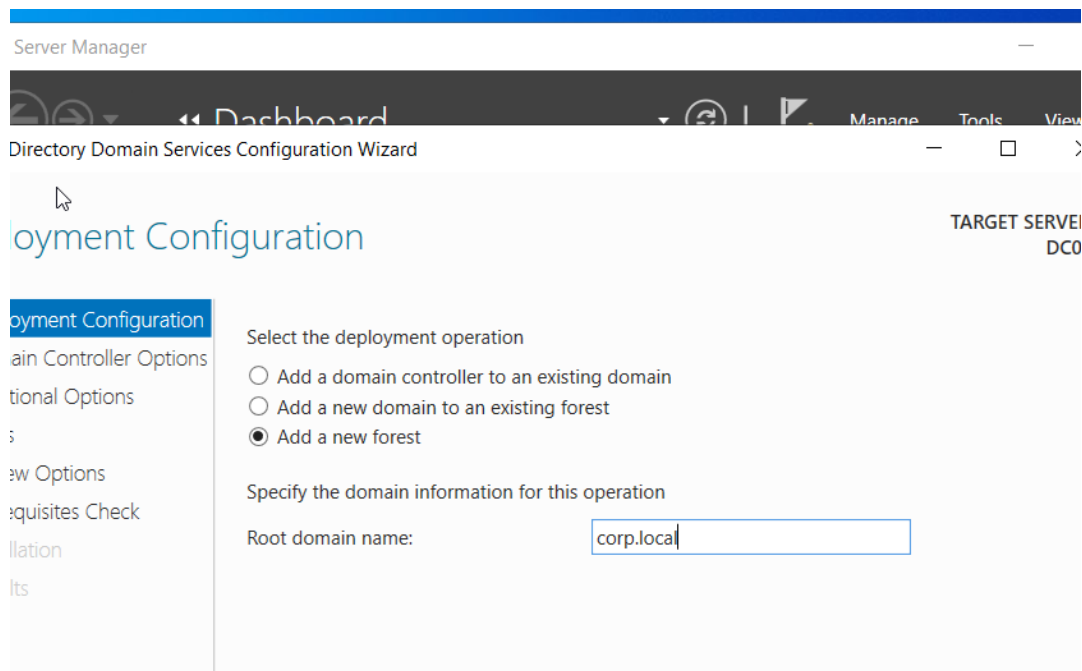
Step 3 – Promote Server to Domain Controller

With the AD DS installation completed, I used the post-deployment configuration wizard to promote the server to a Domain Controller. This step enabled the server to host and manage the Active Directory environment.



Step 4 – Create New Forest

To create a new Active Directory environment, I selected the option to **add a new forest** and configured the root domain name as corp.local. This domain would serve as the foundation for user authentication, computer management, and resource access throughout the lab.



Note: A *forest* is the highest-level logical structure in Active Directory. Creating a new forest establishes a completely new AD environment with its own domain database, security boundaries, and administrative structure.

Step 5 – Configure Domain Controller Options

I configured the Domain Controller options and enabled **DNS Server** and **Global Catalog (GC)**, which are essential components of an Active Directory environment. I also created a **Directory Services Restore Mode (DSRM) password** that can be used for recovery and maintenance tasks if Active Directory needs to be restored in the future.

in Services Configuration Wizard

Controller Options

TARGET SERVER DC01

uration
Options

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016

Domain functional level: Windows Server 2016

Specify domain controller capabilities

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

Note: DSRM (Directory Services Restore Mode) provides administrative access to a domain controller when Active Directory services are offline for maintenance or recovery.

Step 6 – Configure Additional Settings

The default **NetBIOS name (CORP)** was retained, and the standard locations for the **Database**, **Log Files**, and **SYSVOL** folders were accepted. A DNS delegation warning was displayed during the configuration process, which was expected in this isolated lab environment and did not impact the Active Directory deployment.

in Services Configuration Wizard

ptions

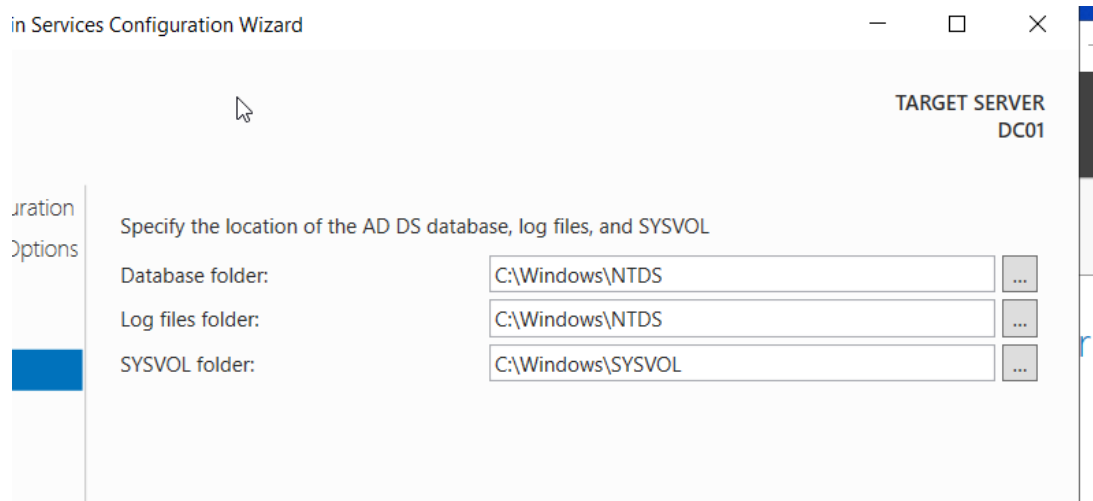
TARGET SERVER DC01

uration
Options

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name: CORP

Note: The wizard automatically generated the **NetBIOS domain name (CORP)** from the root domain **corp.local**. The default value was retained as it follows the domain naming convention used throughout the lab.

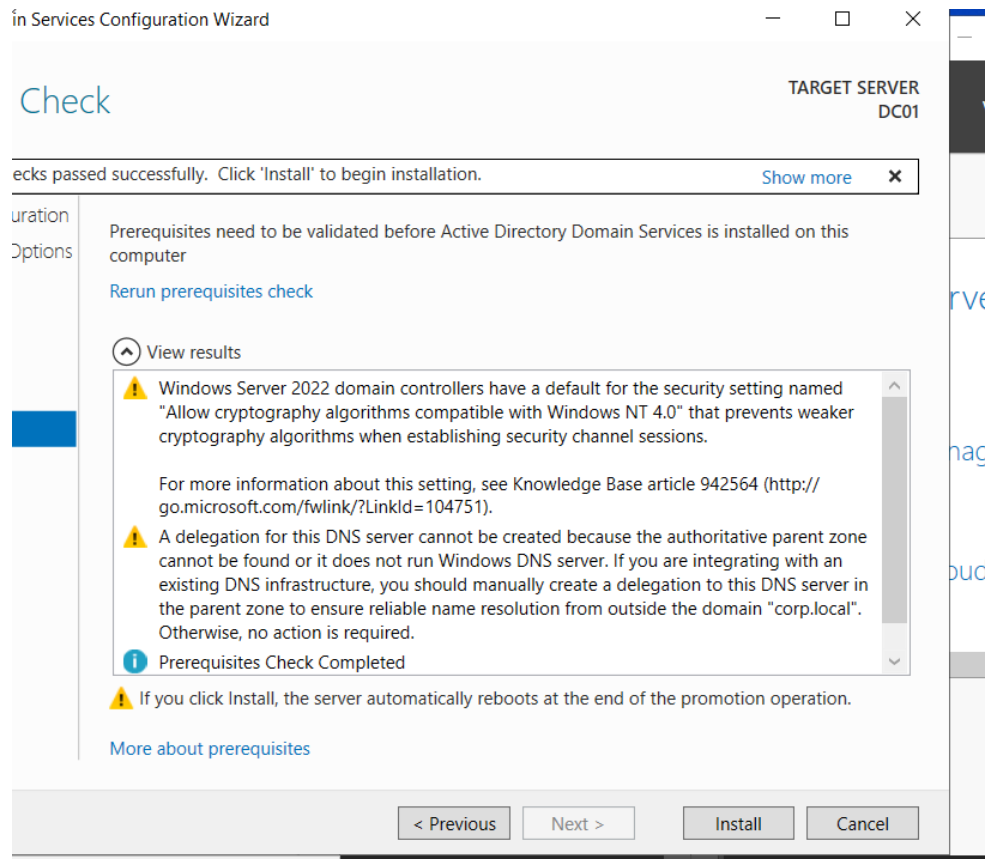


[!NOTE]

The default locations for the **AD DS database**, **log files**, and **SYSVOL** folder were reviewed and retained. These locations are suitable for a lab environment and support Active Directory and Group Policy operations.

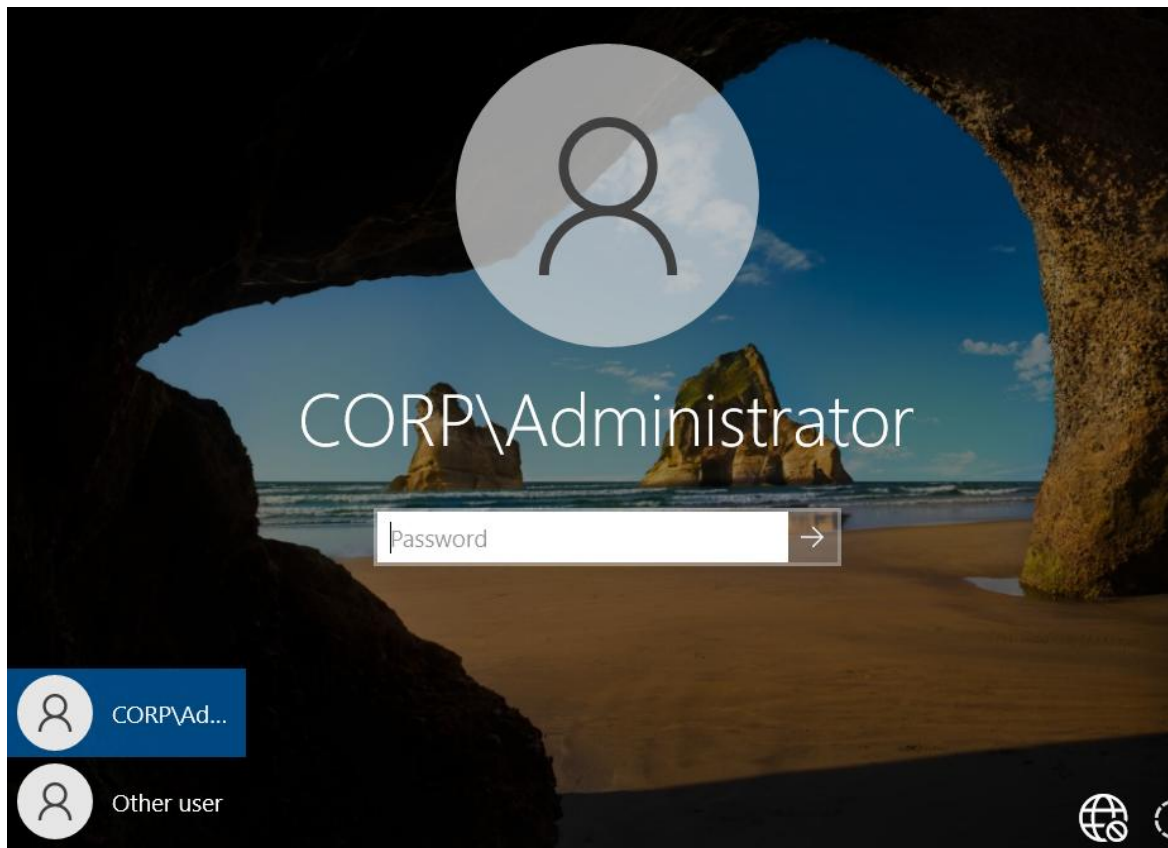
Step 7 – Perform Prerequisite Check

I reviewed the prerequisite check results and confirmed that all required conditions were met before proceeding with the Domain Controller installation.



Step 8 – Install and Promote

After clicking **Install**, the server completed the Domain Controller promotion process and automatically restarted. Following the reboot, the login account changed from the local **Administrator** account to **CORP\Administrator**, confirming that the server had successfully joined the newly created Active Directory domain and was now operating as a Domain Controller.



Phase Summary

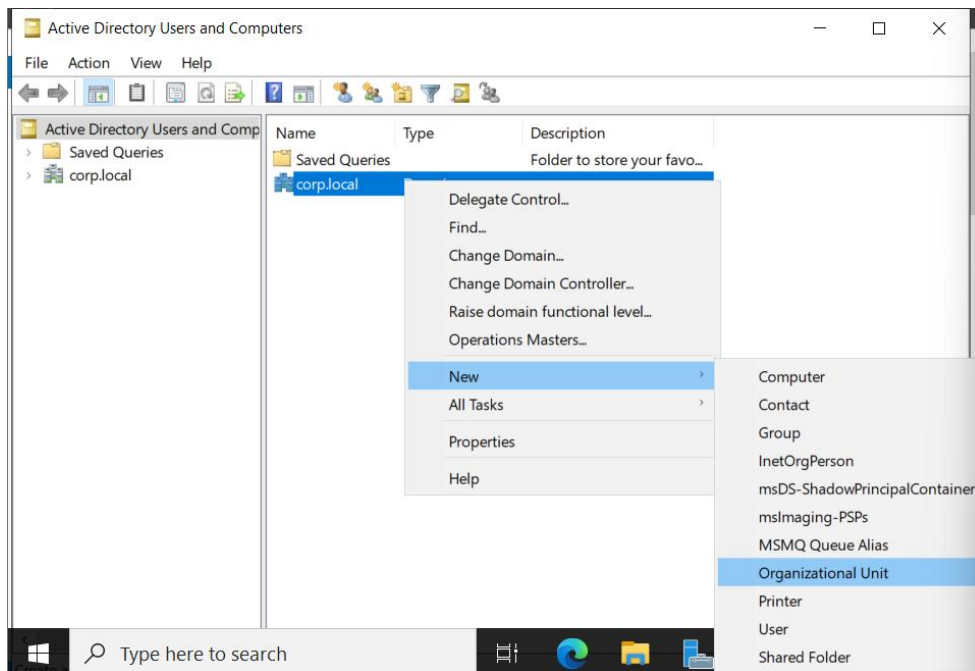
In this phase, I installed Active Directory Domain Services and promoted the server to a Domain Controller. A new Active Directory forest (**corp.local**) was successfully created, providing the foundation for user, computer, and policy management throughout the lab environment.

Phase 3 – Create Organizational Units

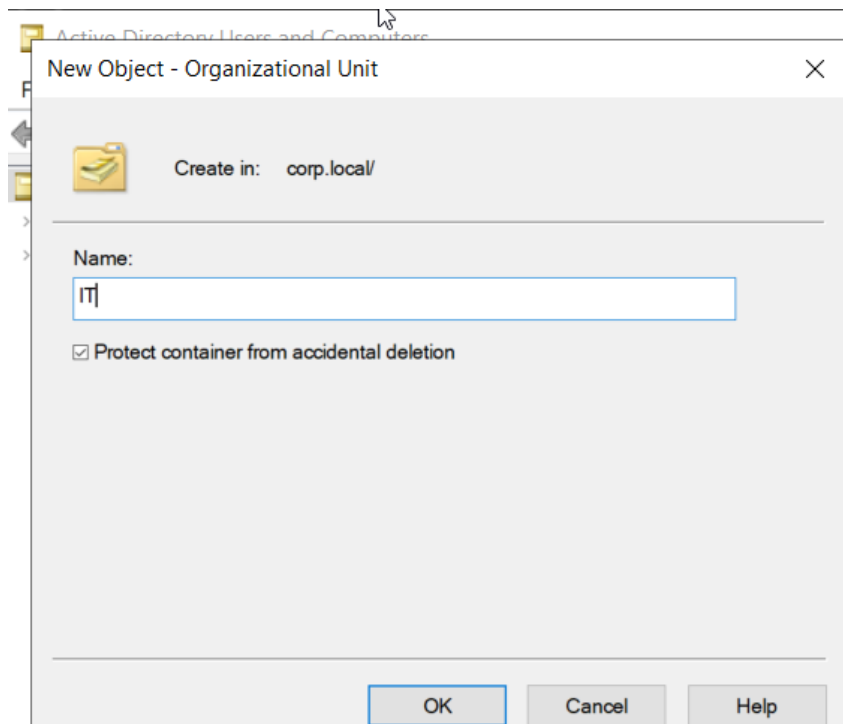
Overview

In this phase, I created Organizational Units (OUs) to establish a logical structure for managing users, groups, and devices within the Active Directory environment. Organizing resources into separate OUs makes administration easier and supports the future application of Group Policies.

Using Active Directory Users and Computers, I selected the **Organizational Unit** option to create dedicated containers for different departments and administrative purposes within the domain.

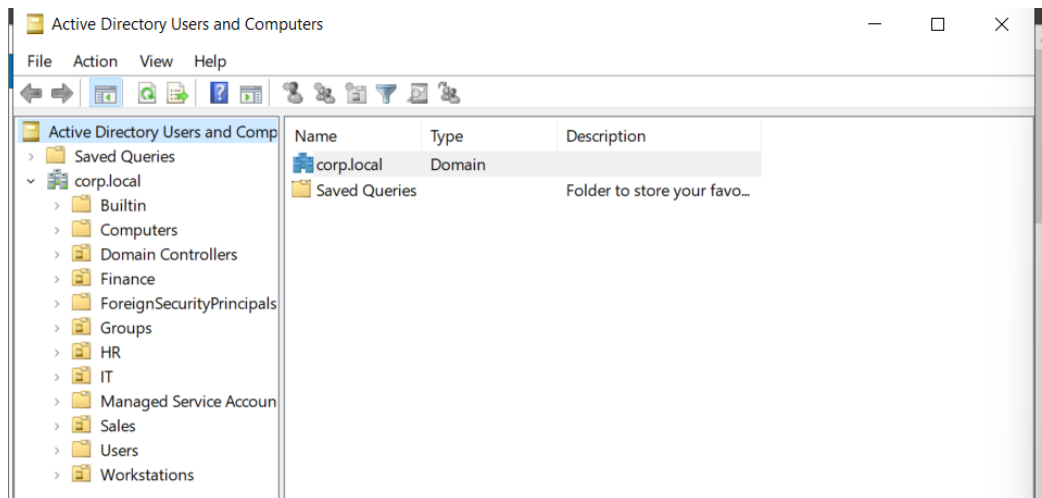


Each Organizational Unit was created with a meaningful name based on its intended purpose. This structure helps keep Active Directory organized and simplifies the management of users, computers, and security policies.



The Organizational Unit structure was successfully created within the **corp.local** domain. Separate OUs were configured for departments such as **IT**, **HR**, **Finance**, and

Sales, providing a clear administrative hierarchy for future user and policy management.



Phase Summary

A structured Organizational Unit hierarchy was successfully created within the Active Directory environment. This provided a foundation for organizing users, groups, and devices while supporting more efficient administration and future Group Policy deployment.

Phase 4- Create Users and Security Admin

Overview

In this phase, I created user accounts and security groups within Active Directory to simulate a departmental access control structure. Users were assigned to appropriate security groups to demonstrate Role-Based Access Control (RBAC), allowing permissions to be managed through group membership rather than individual user accounts.

Steps Performed

I created a user account for **John Smith** within the **IT** Organizational Unit and configured the account with a domain logon name. This simulated the onboarding process of a new employee in an Active Directory environment.

Dashboard

Manage

New Object - User

Create in: corp.local/IT

First name: John Initials:

Last name: Smith

Full name: John Smith

User logon name: john.smith @corp.local

User logon name (pre-Windows 2000): CORP\ john.smith

< Back Next > Cancel

To support role-based access management, I created a **Security Group** named **IT-Users** using the **Global** group scope. Security groups provide a scalable way to assign permissions to multiple users based on their job role or department.

New Object - Group

Create in: corp.local/Groups

Group name: IT-Users

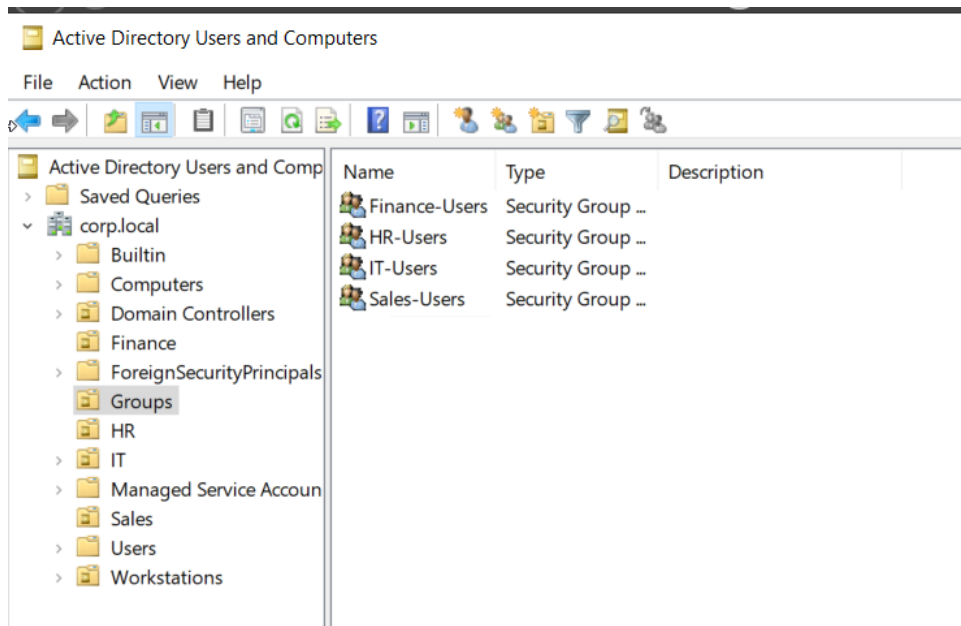
Group name (pre-Windows 2000): IT-Users

Group scope: ☐ Domain local ☒ Global ☐ Universal

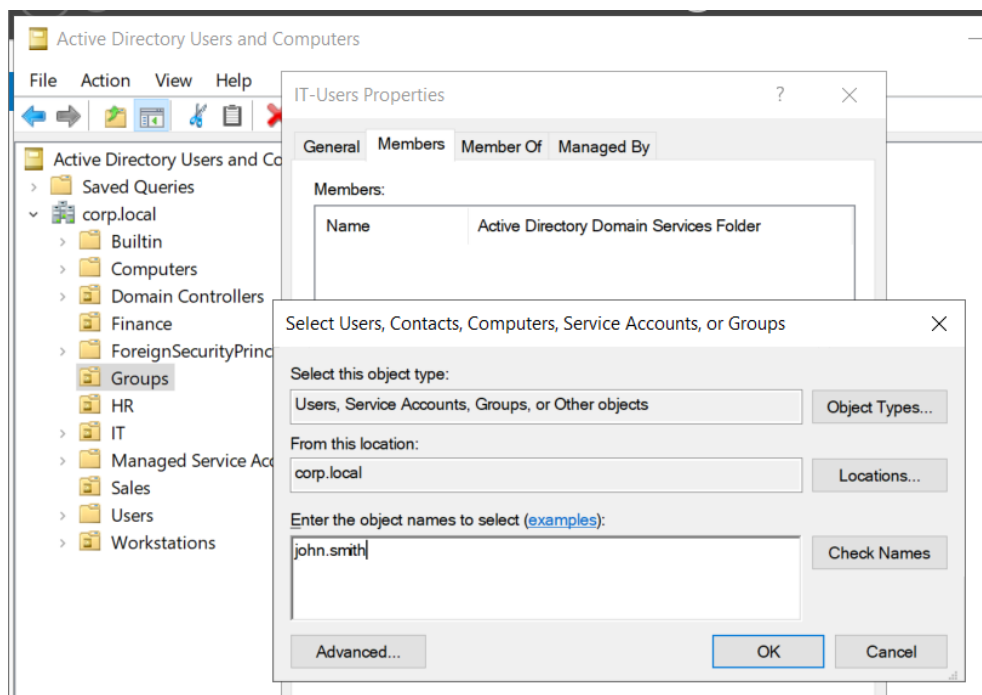
Group type: ☒ Security ☐ Distribution

OK Cancel

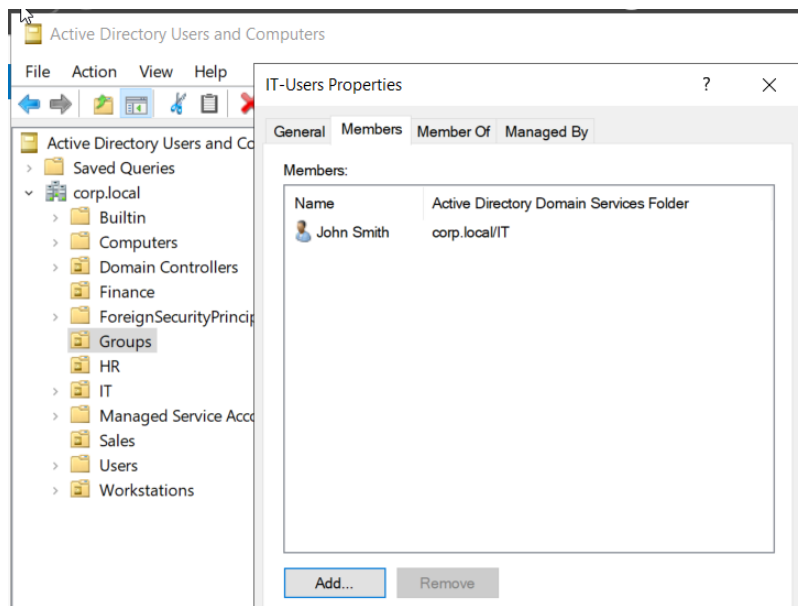
Departmental security groups were created for **IT**, **HR**, **Finance**, and **Sales**. This structure allows administrators to manage access rights more efficiently by assigning permissions to groups rather than individual user accounts.



I added **John Smith** to the **IT-Users** security group, allowing access permissions to be inherited through group membership. This follows the RBAC principle commonly used in enterprise environments.



The user's group memberships were verified to confirm that **John Smith** was successfully assigned to **IT-Users** while remaining a member of the default **Domain Users** group. This validated that the RBAC configuration had been applied correctly.



Phase Summary

User accounts and departmental security groups were successfully created and configured within Active Directory. Group memberships were verified, demonstrating how Role-Based Access Control (RBAC) can simplify user administration and permission management in a Windows domain environment.

Phase 5 – Create Shared Folder and Assign Permissions

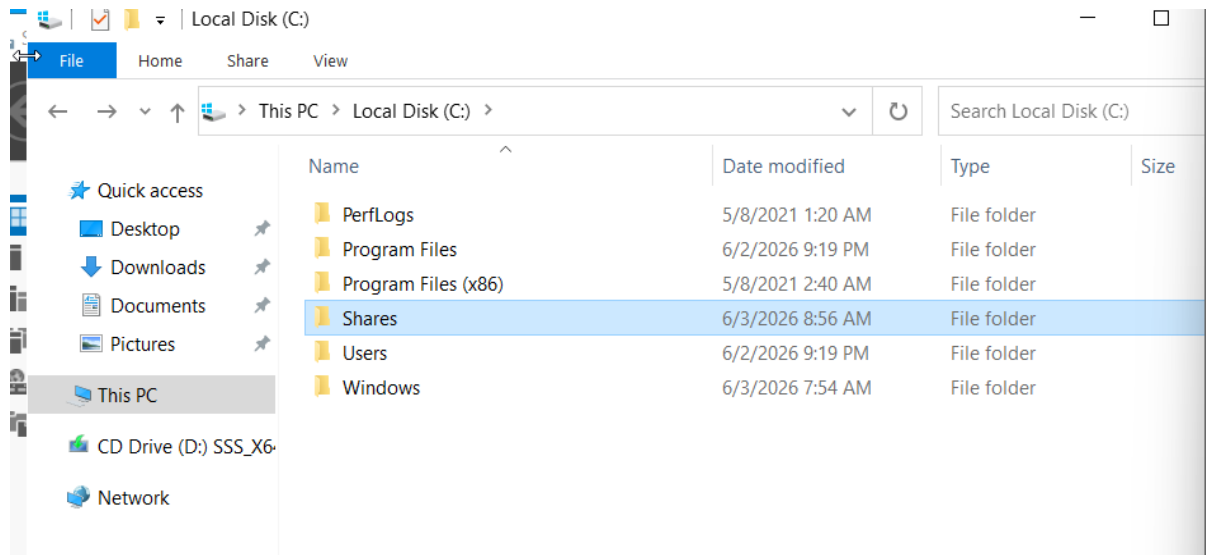
Overview

In this phase, I created a shared folder on Windows Server and configured access permissions using Active Directory security groups. This demonstrates how Role-Based Access Control (RBAC) can be used to manage departmental resources in a centralized domain environment.

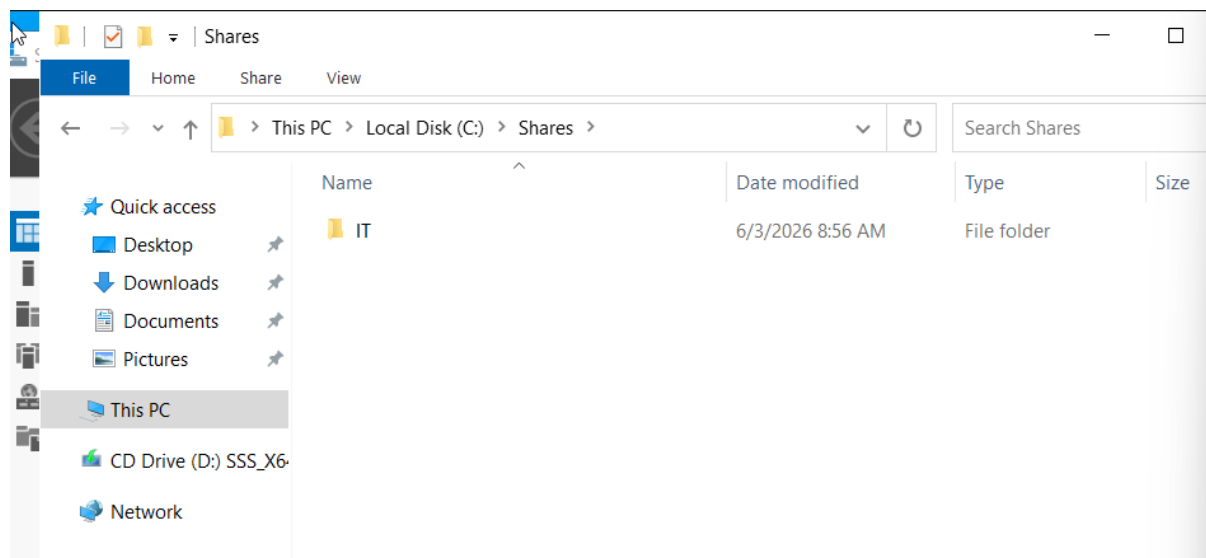
Steps Performed

Step 1 – Create Shared Folder

To simulate a departmental file share, I created a folder structure under **C:\Shares** and configured a dedicated IT folder. This directory would later be shared across the network and secured using Active Directory security groups to control access based on user roles.

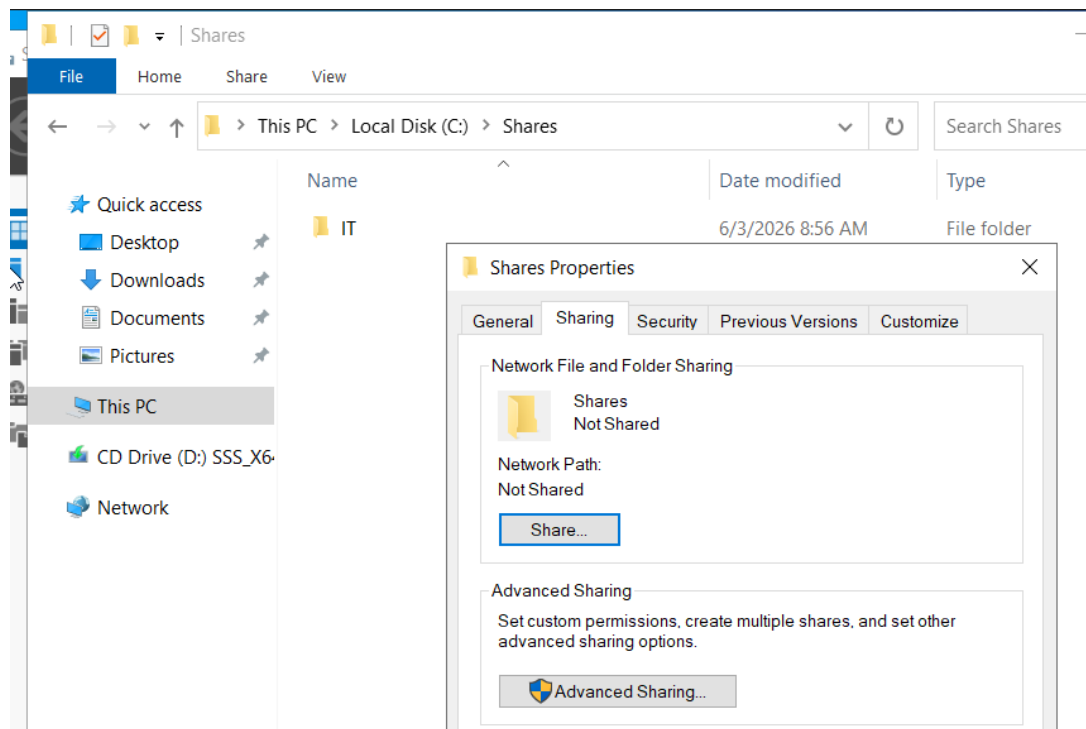


Within the Shares directory, I created an **IT** folder that would be used as a departmental file share. Access to this folder would later be controlled through Active Directory security groups.



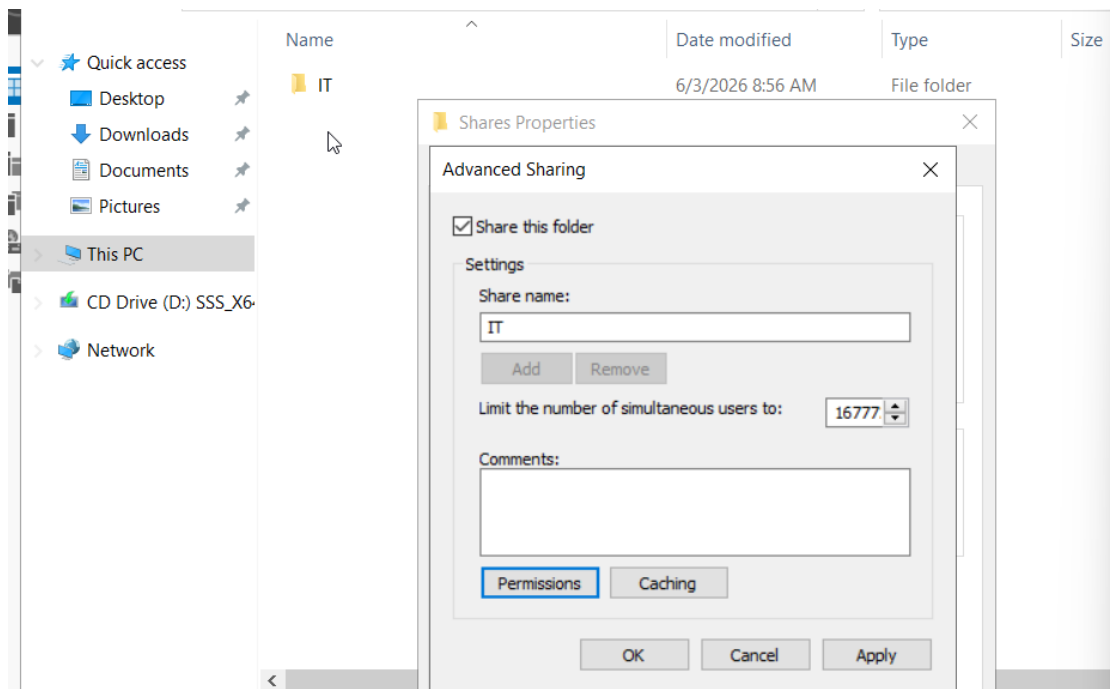
Step 2 – Configure Folder Sharing

To make the folder accessible across the network, I configured sharing settings for the **IT** directory using **Advanced Sharing**. The folder was published as a network share named **IT**, allowing authorized domain users to access it through the path **\\DC01\\IT**.

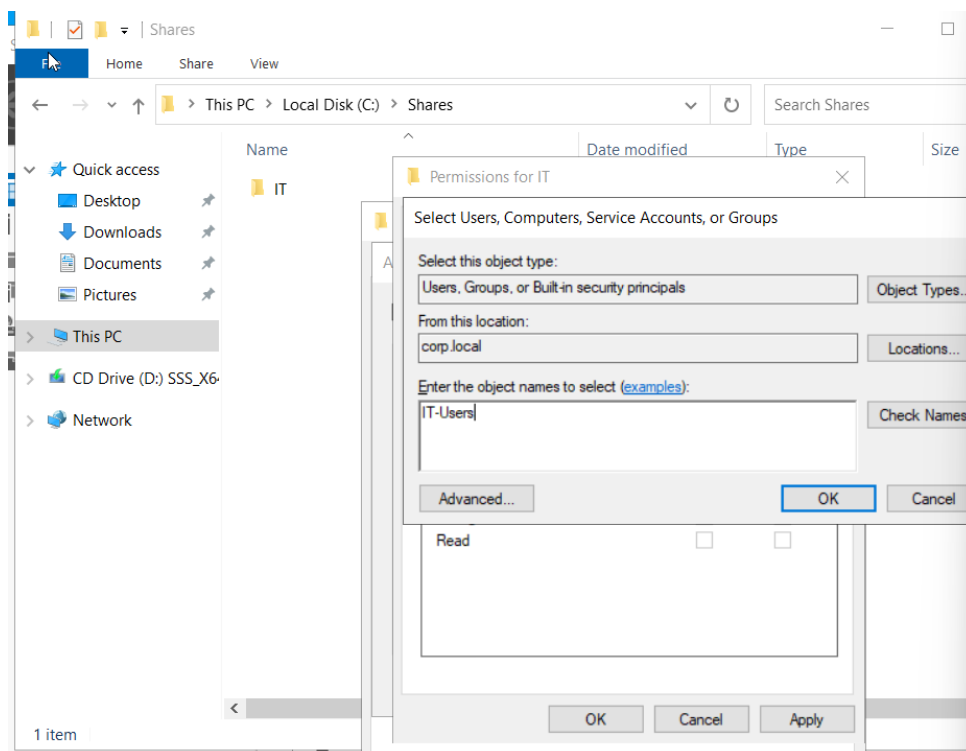


Step 3 – Assign Share Permissions

After enabling folder sharing, I configured share-level permissions to control which Active Directory groups could access the resource. Using security groups simplifies administration and ensures permissions can be managed centrally.

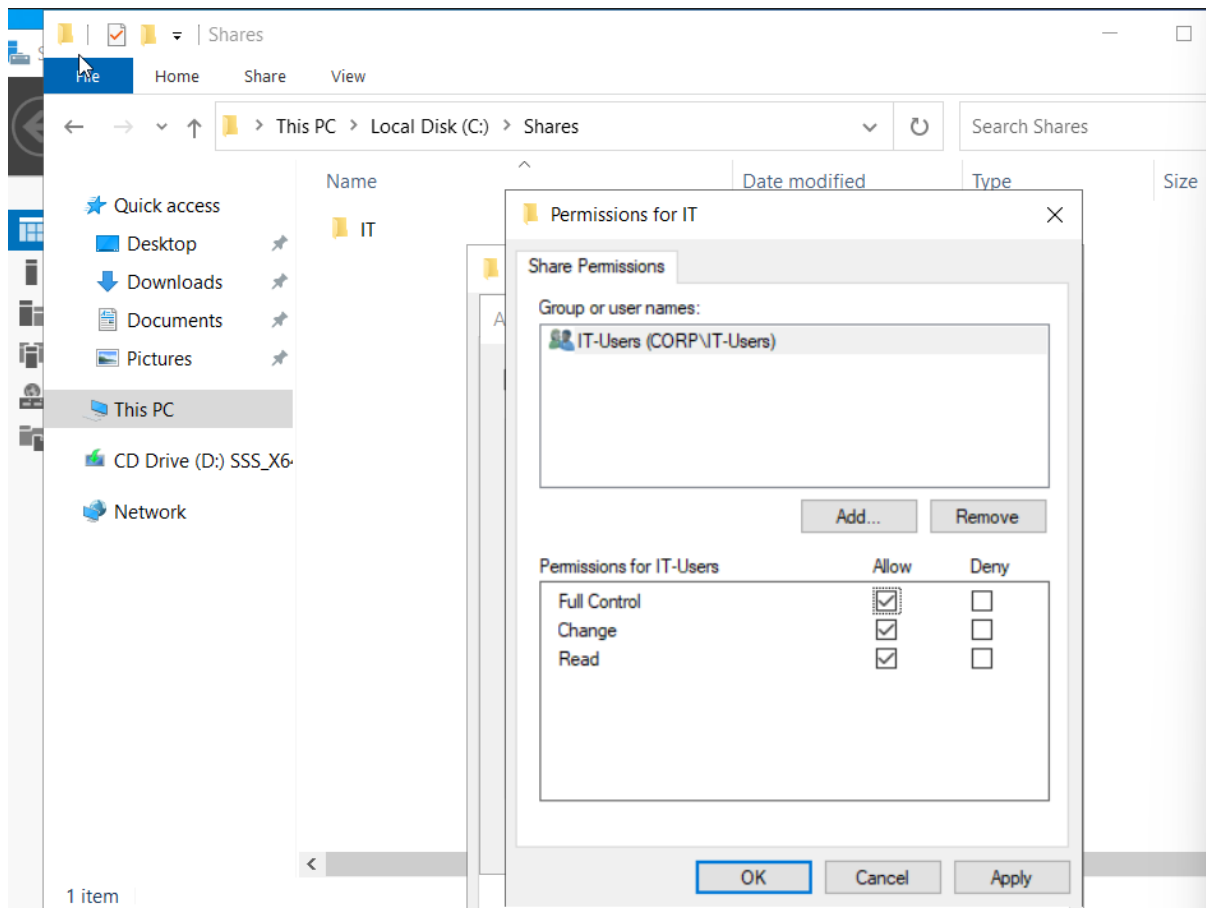


The **IT-Users** security group was added to the shared folder permissions. This approach allows access to be granted based on group membership rather than assigning permissions to individual user accounts.



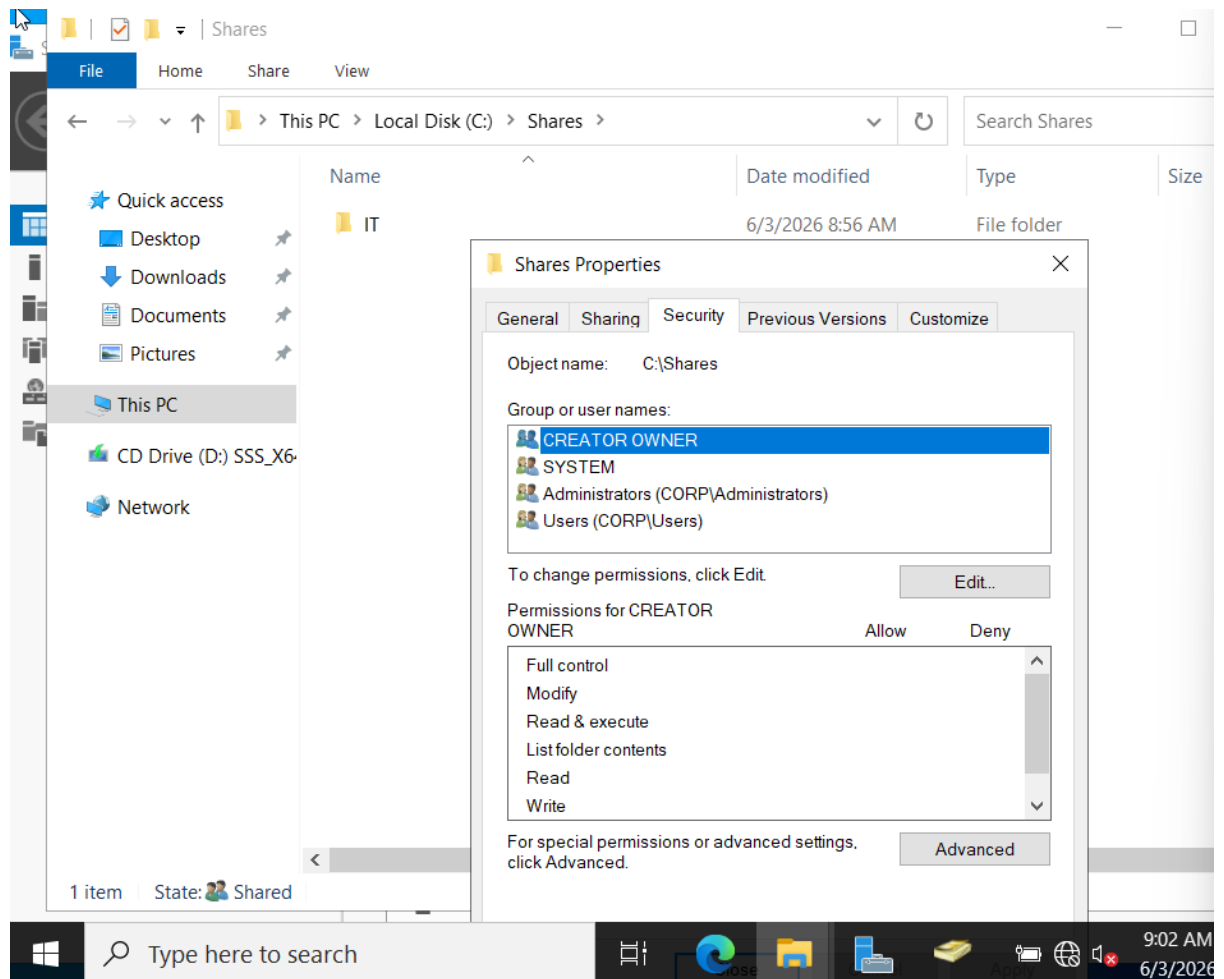
Step 4 – Configure Share Permissions

The IT-Users security group was granted **Full Control** permissions on the shared folder. Access was assigned through group membership rather than individual user accounts, following the Role-Based Access Control (RBAC) model commonly used in enterprise environments.



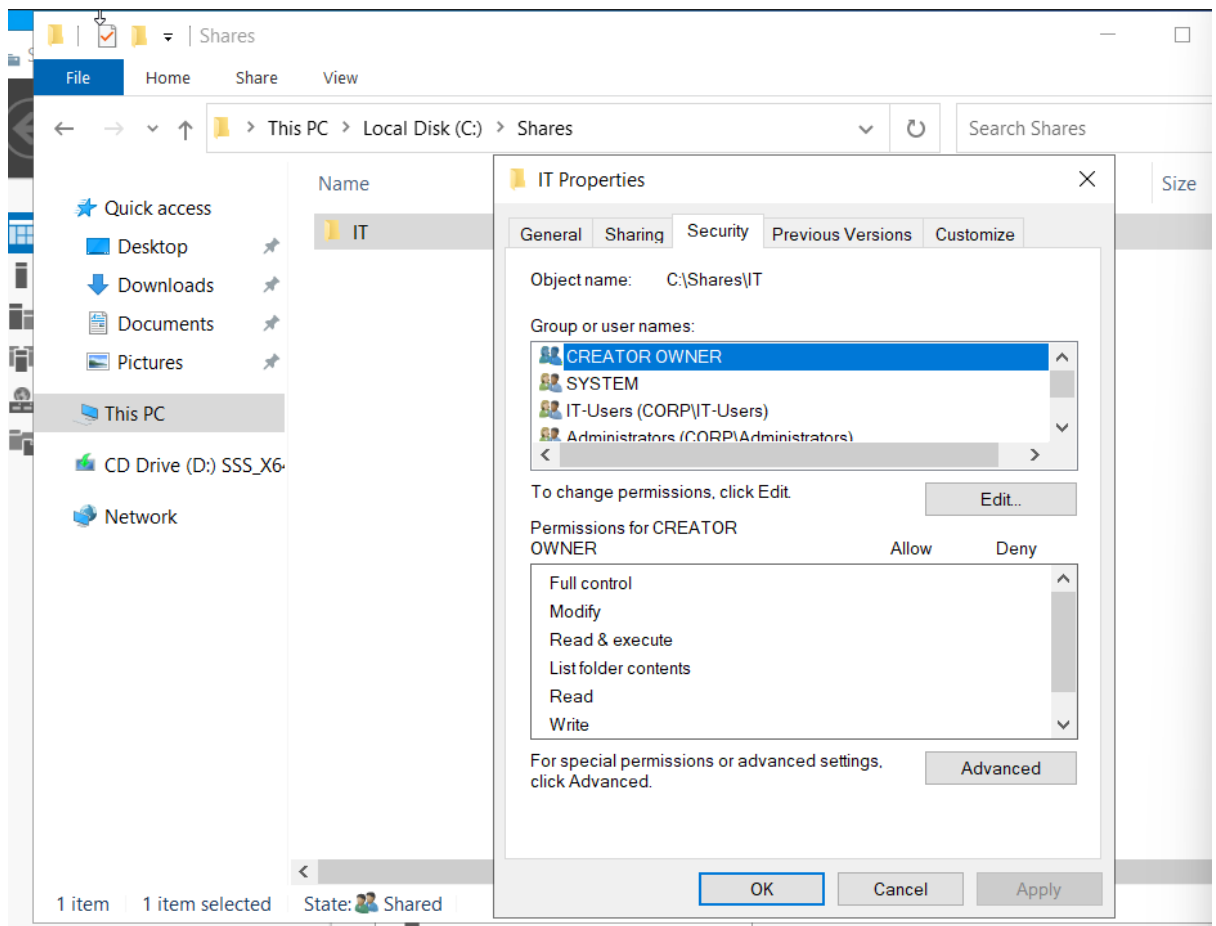
Step 5 – Configure NTFS Permissions

After configuring share-level permissions, I reviewed the **NTFS** security settings of the folder. NTFS permissions provide an additional layer of access control and help ensure users receive only the permissions required for their role.



Note: NTFS (New Technology File System) permissions provide file and folder-level security in Windows. They allow administrators to control who can access, modify, or manage data stored on a server.

The IT-Users security group was added to the folder's NTFS permissions and granted the appropriate access rights. By combining Share Permissions and NTFS Permissions, access to the departmental file share can be managed securely and centrally through Active Directory groups.



Phase Summary

A departmental file share was successfully created and published on the network. Share Permissions and NTFS Permissions were configured using Active Directory security groups, demonstrating how Role-Based Access Control (RBAC) can be used to manage access to shared resources in a Windows Server environment.

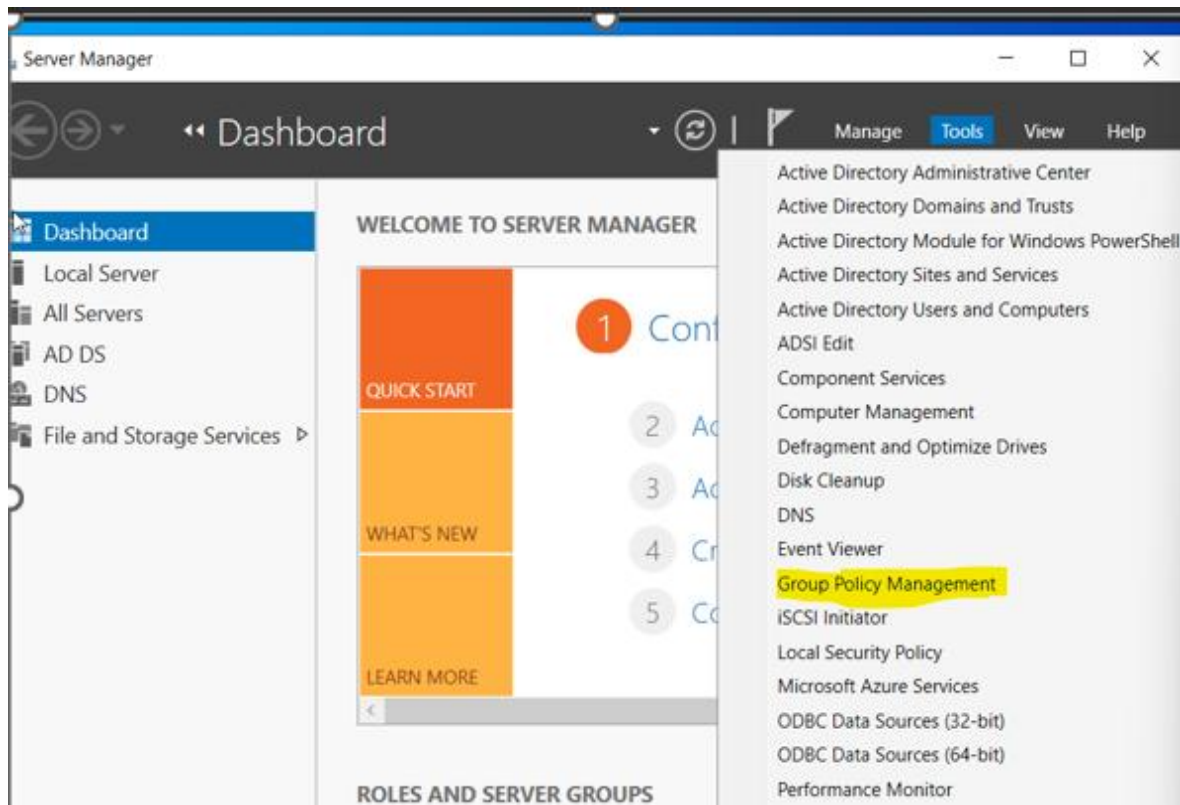
Phase 6 – Configure Password and Account Lockout Policies

Overview

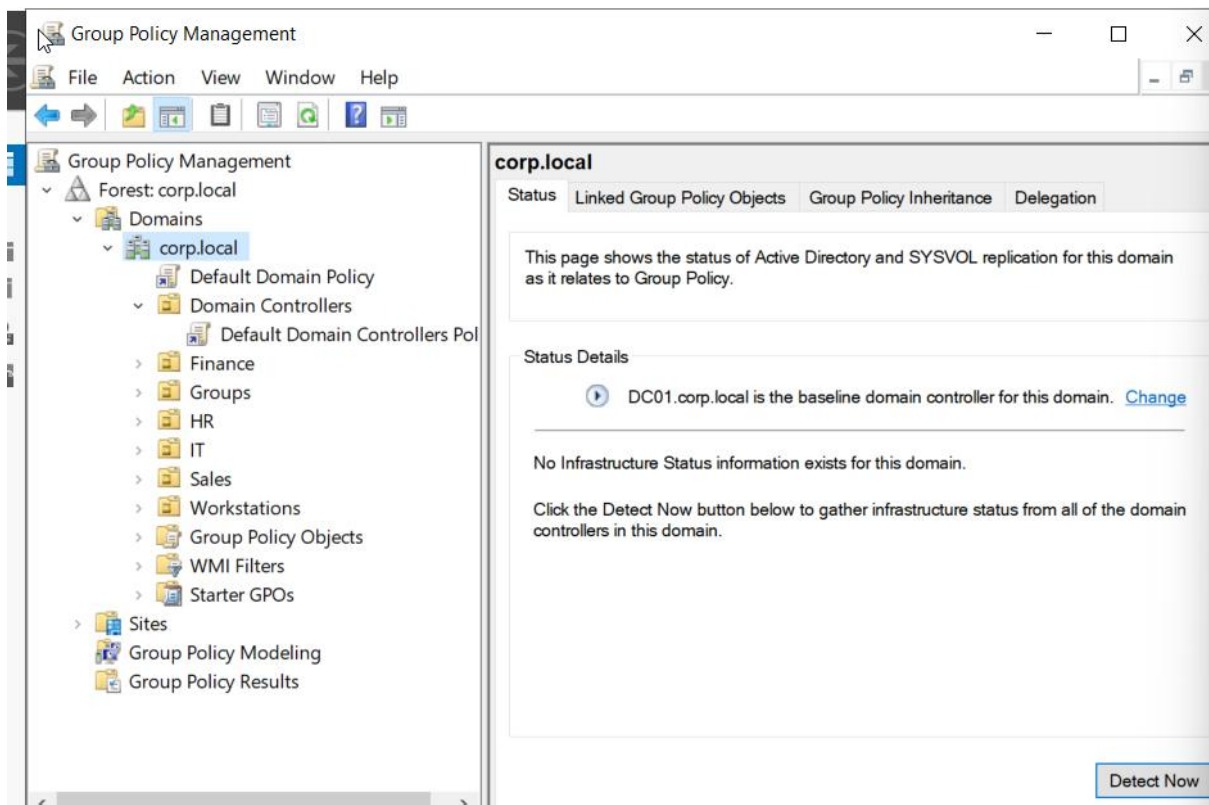
This phase focuses on improving domain security by configuring password and account lockout policies through Group Policy Management. These settings ensure that all domain users follow consistent password requirements and help reduce the risk of unauthorized access.

Step 1. Open Group Policy Management

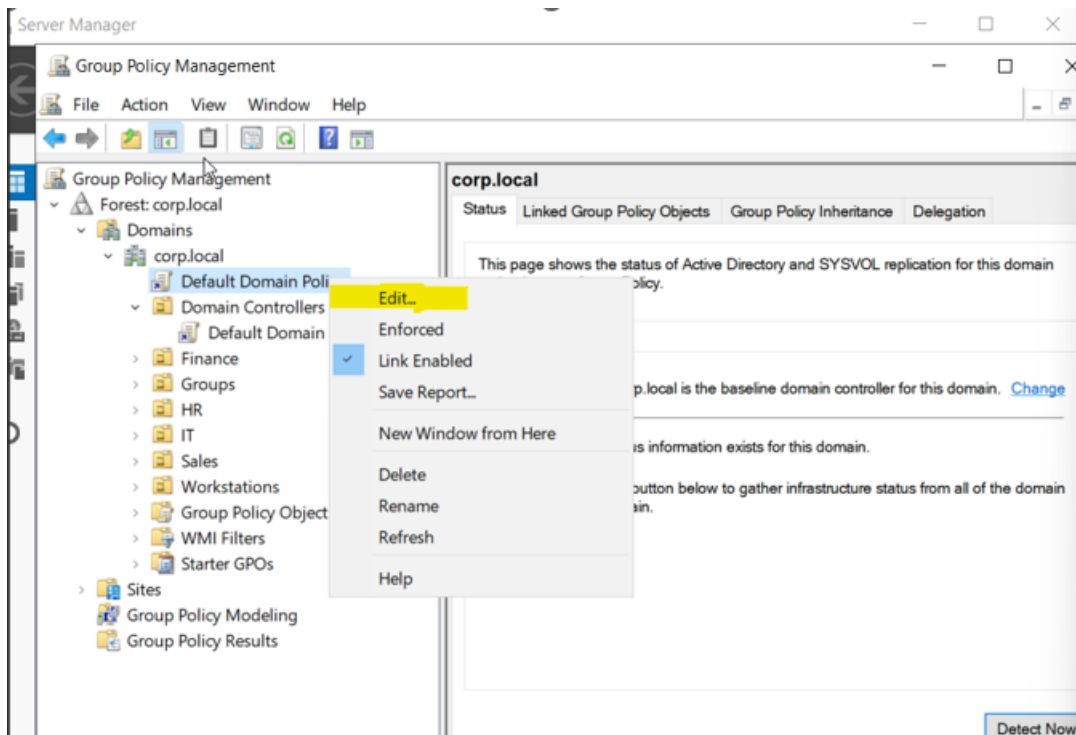
I launched **Group Policy Management** Console (GPMC) from Server Manager to configure domain-wide security policies. GPMC provides centralized management of Group Policy Objects (GPOs) across the Active Directory environment.



Within the **corp.local** domain, I navigated to the **Default Domain Policy**. This policy applies to all domain users and computers and is commonly used to enforce organization-wide security settings.



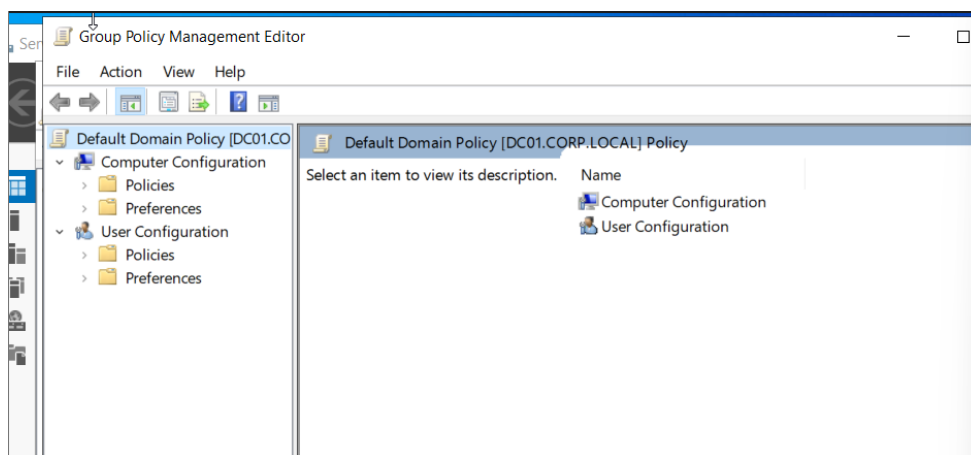
The Default Domain Policy was opened for editing to configure password and account lockout settings. These policies help strengthen authentication security and reduce the risk of unauthorized access.



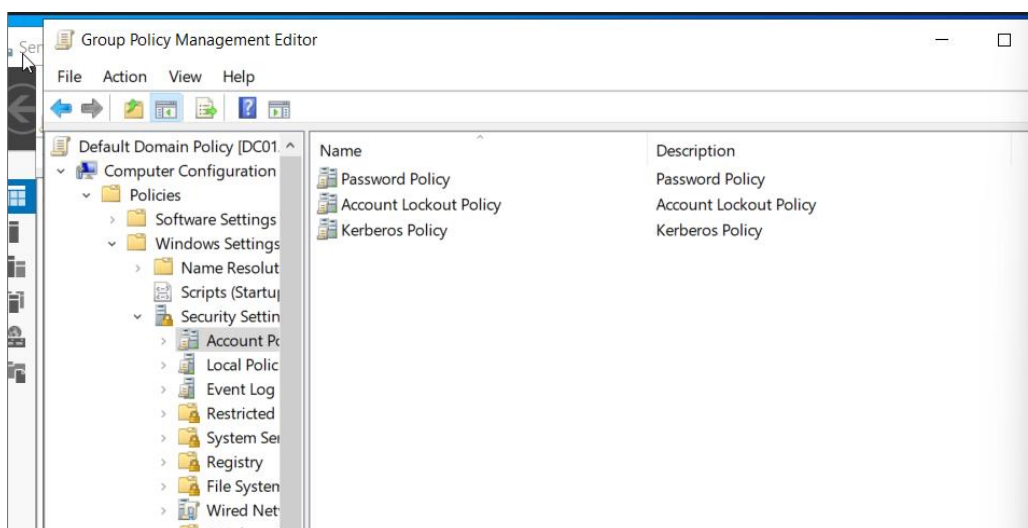
Note: Domain-level password policies are managed through Group Policy and apply to all users within the Active Directory domain unless fine-grained password policies are implemented.

Step 2. Configure Password Policy

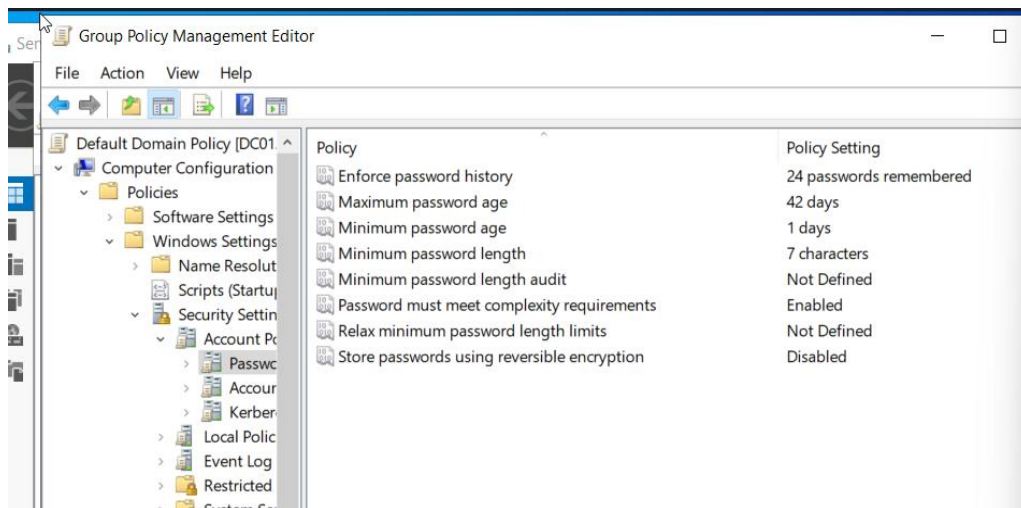
Opened **Group Policy Management Editor** and accessed the **Default Domain Policy**. This policy is applied to all users and computers within the Active Directory domain, making it the standard location for configuring domain-wide password requirements.



Navigated through the Group Policy structure to locate the **Password Policy** settings under **Account Policies**. This section contains the rules that control password requirements for all users in the domain.



Reviewed the existing password policy and verified the security settings, including password length, password complexity requirements, password history, and password expiration settings. These controls help reduce the risk of weak passwords and improve overall domain security.

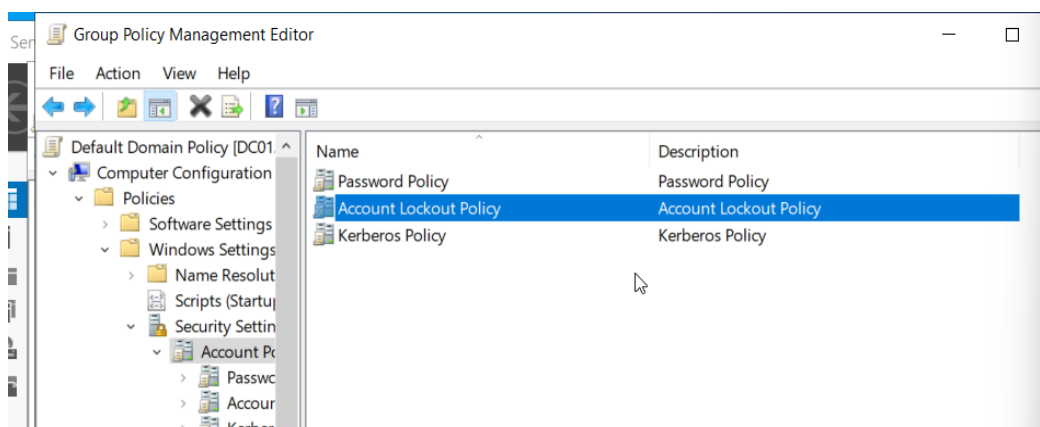


Note:

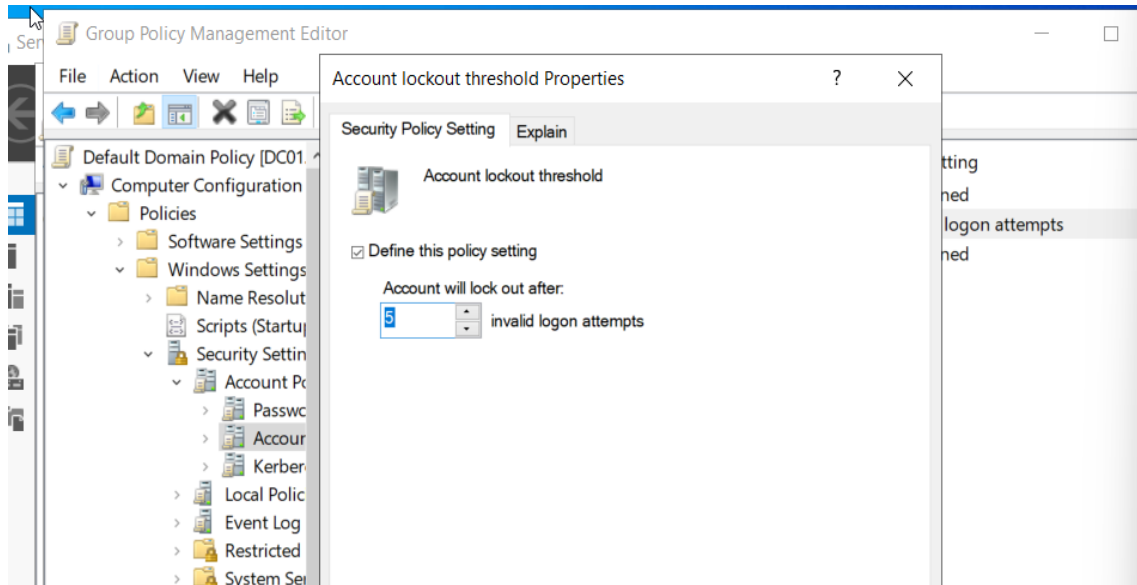
Changes made in the **Default Domain Policy** are automatically applied to all Active Directory user accounts within the **corp.local** domain after Group Policy refresh.

Step 3. Configure Account Lockout Policy

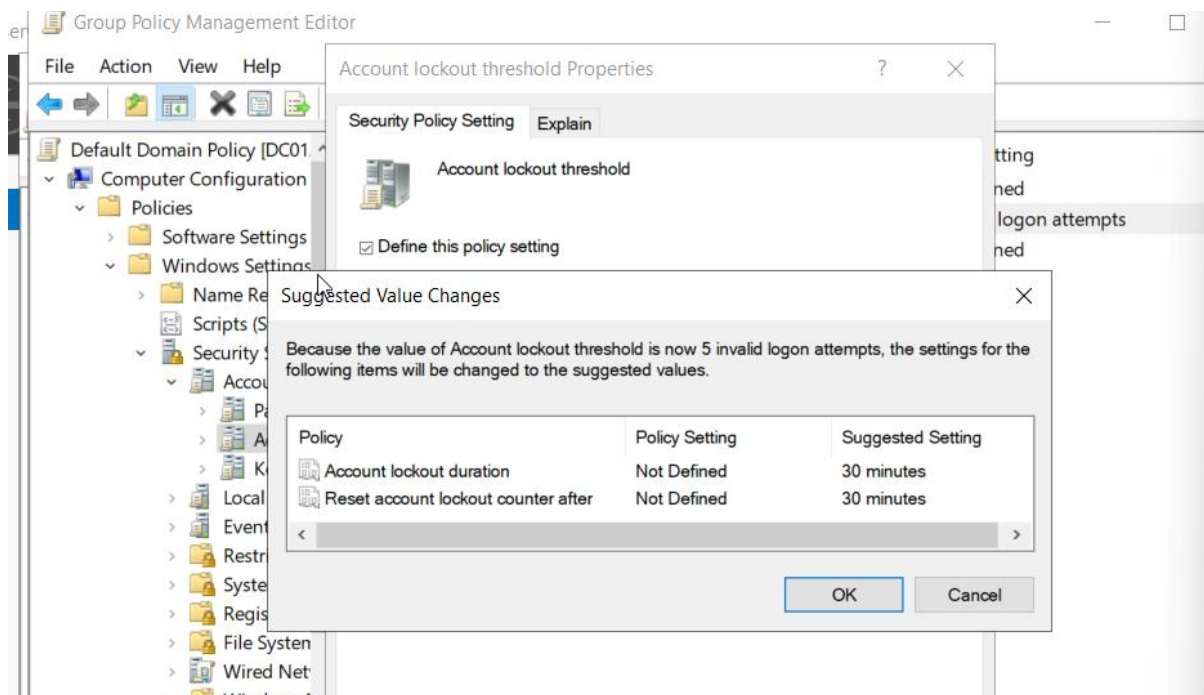
Opened the **Account Lockout Policy** section within the Default Domain Policy. This policy controls how Active Directory responds to repeated failed sign-in attempts and helps protect user accounts from unauthorized access.



Configured the **Account Lockout Threshold** to lock a user account after **5 invalid logon attempts**. This setting helps reduce the risk of brute-force attacks by preventing unlimited password guesses.



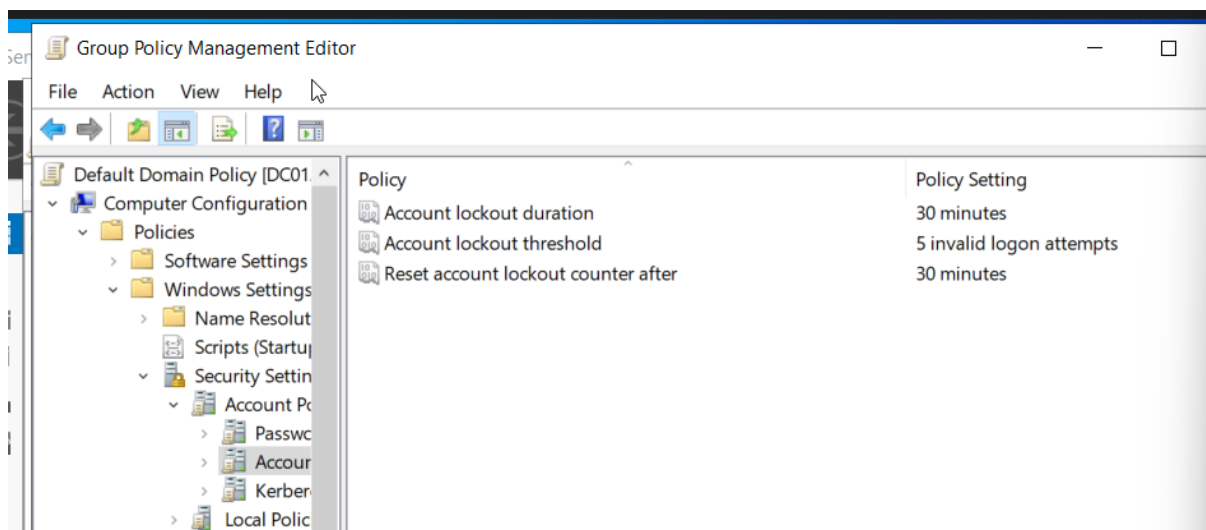
Accepted the recommended settings generated by Windows. The lockout duration and reset counter values were automatically set to **30 minutes**, providing a balance between security and user convenience.



Verified that the Account Lockout Policy was successfully configured with the following settings:

- Account lockout threshold: 5 invalid logon attempts
- Account lockout duration: 30 minutes
- Reset account lockout counter after: 30 minutes

These settings ensure that accounts are temporarily locked after multiple failed login attempts, helping to protect the domain from password-based attacks.



Note: Account Lockout Policy is a common security control in enterprise environments and is used to limit repeated login attempts against Active Directory user accounts.

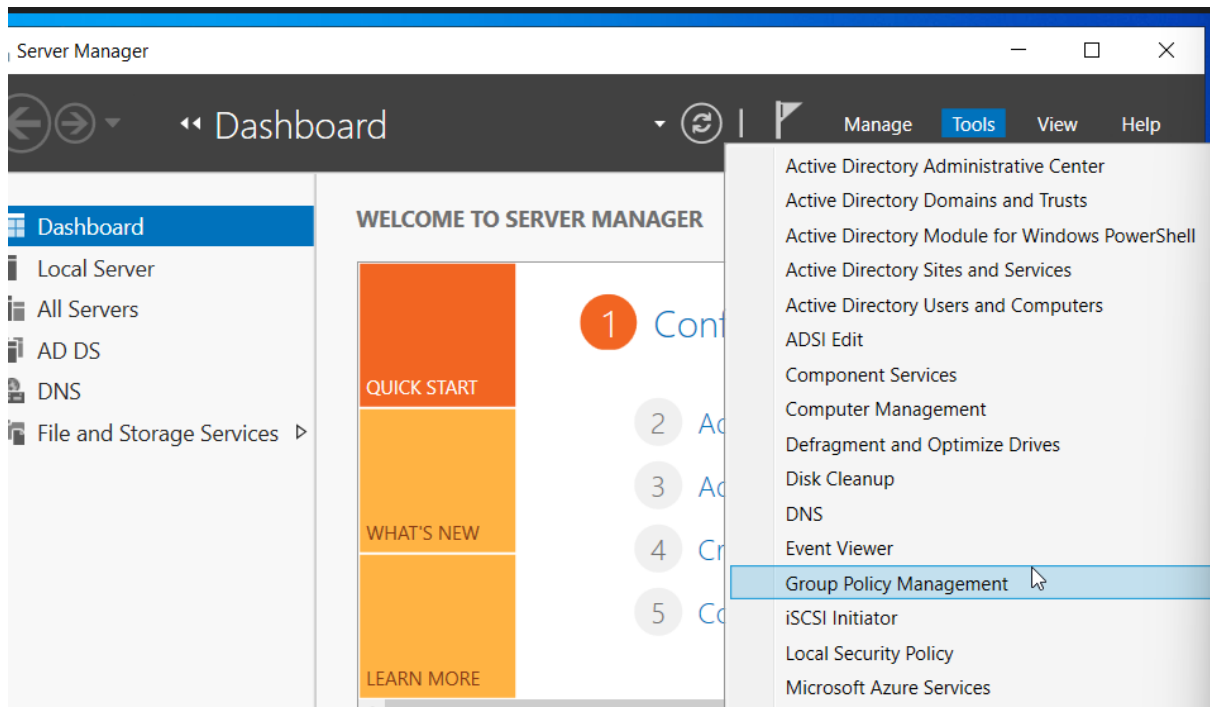
Phase 7 – Create and Link a Custom Group Policy Object (GPO)

Objective

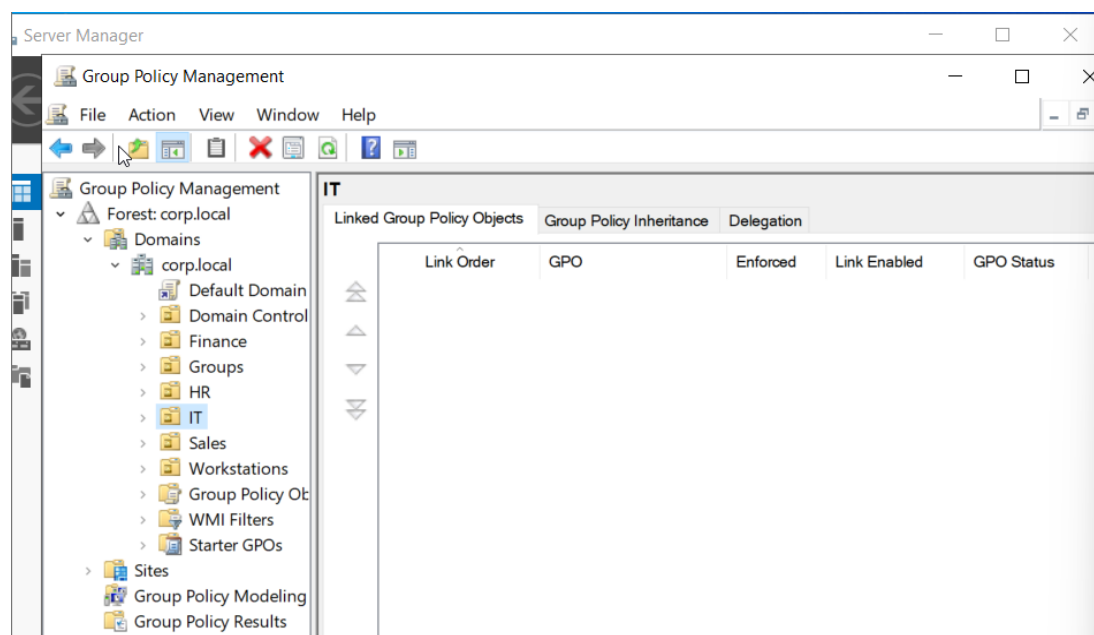
Create and deploy a custom Group Policy Object (GPO) to the IT Organizational Unit (OU) in Active Directory. The policy was designed to restrict access to administrative tools and improve endpoint security for IT department users.

Step 1. Create a New GPO

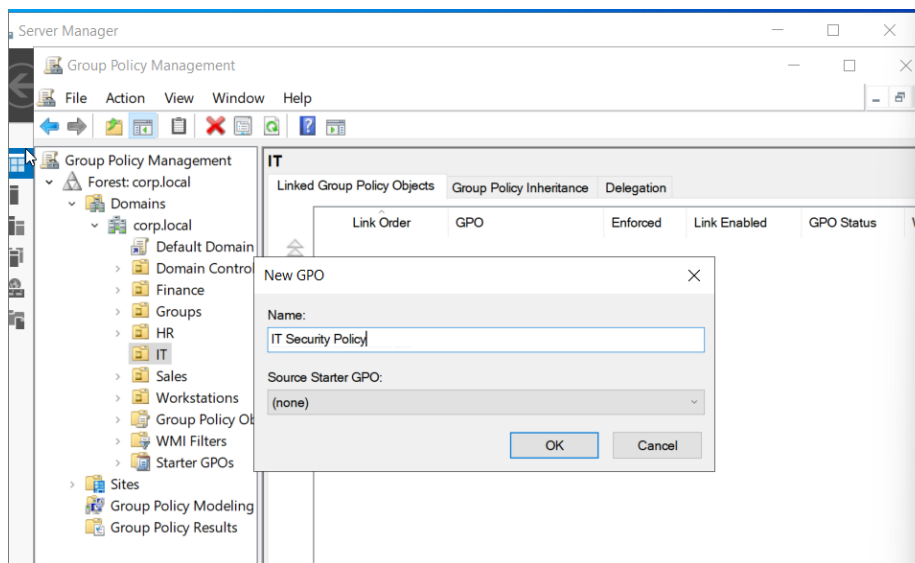
Opened **Group Policy Management** from the Server Manager Tools menu. This console is used to create, manage, and deploy Group Policy Objects (GPOs) across the Active Directory environment.



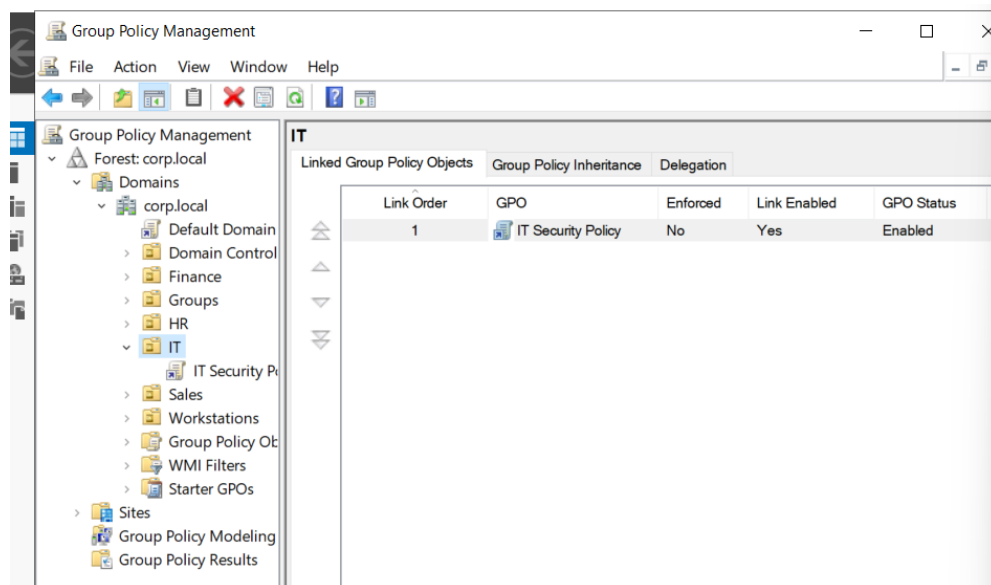
Expanded the domain structure and selected the **IT Organizational Unit (OU)**. The new policy would be linked to this OU so that only IT department users would receive the configured settings.



Created a new Group Policy Object named **IT Security Policy**. Using a separate GPO makes it easier to manage security settings without modifying existing default policies.



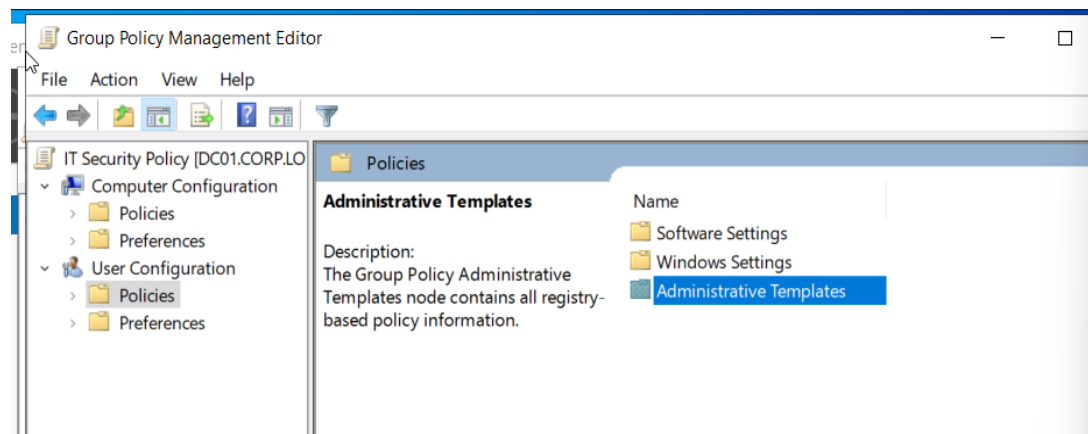
Confirmed that the **IT Security Policy** GPO was successfully linked to the IT Organizational Unit. The policy is now enabled and ready for configuration and deployment.



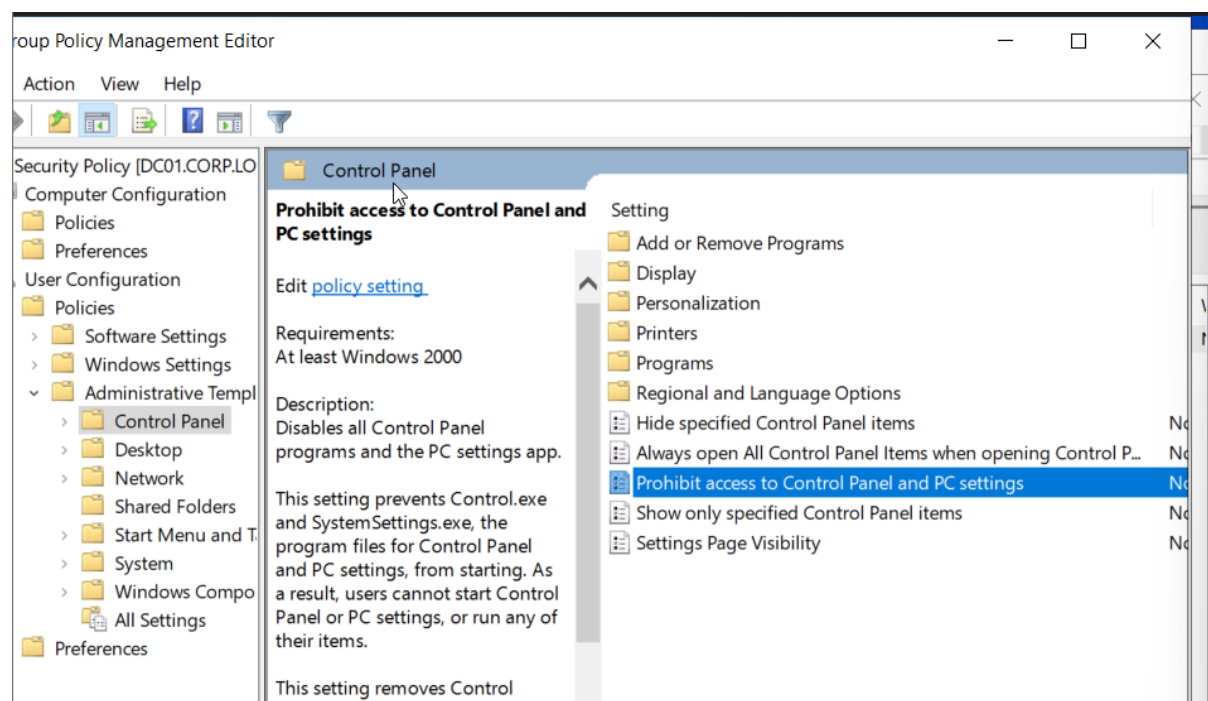
Note: Creating separate GPOs for different departments is a common administrative practice because it allows policies to be managed independently and applied only to the required users or computers.

Step 2. Configure Control Panel Restrictions

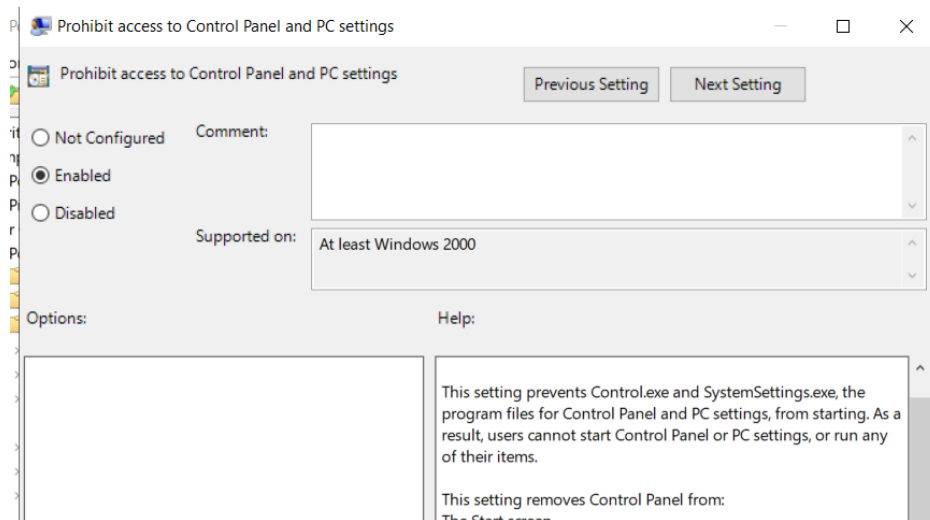
Opened the **Administrative Templates** section in the **IT Security Policy** GPO. This area contains user and computer policy settings that can be applied across the domain.



Navigated to the **Control Panel** policy settings and located the **Prohibit access to Control Panel and PC settings** policy. This setting is used to restrict users from accessing system configuration tools.

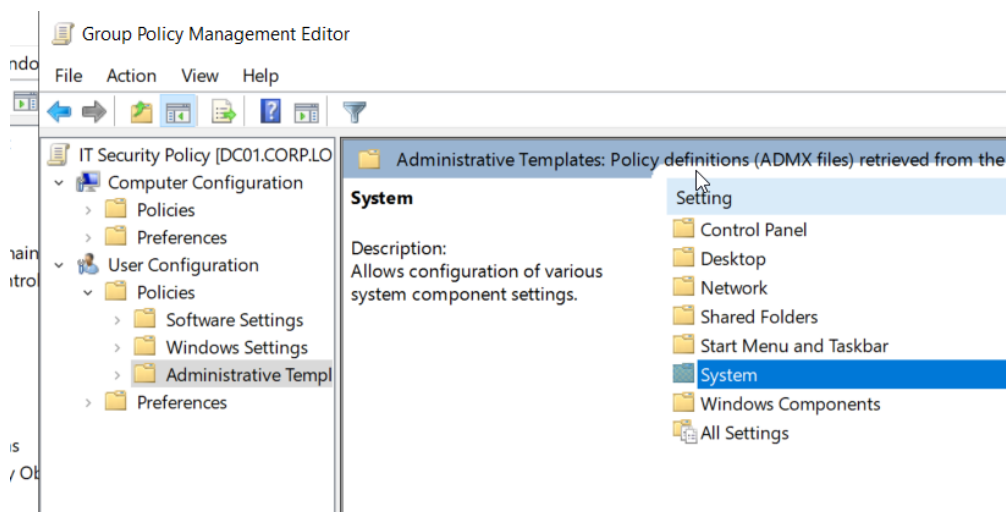


Enabled the **Prohibit access to Control Panel and PC settings** policy. This ensures that users within the IT OU cannot open Control Panel or Windows Settings after the policy is applied.

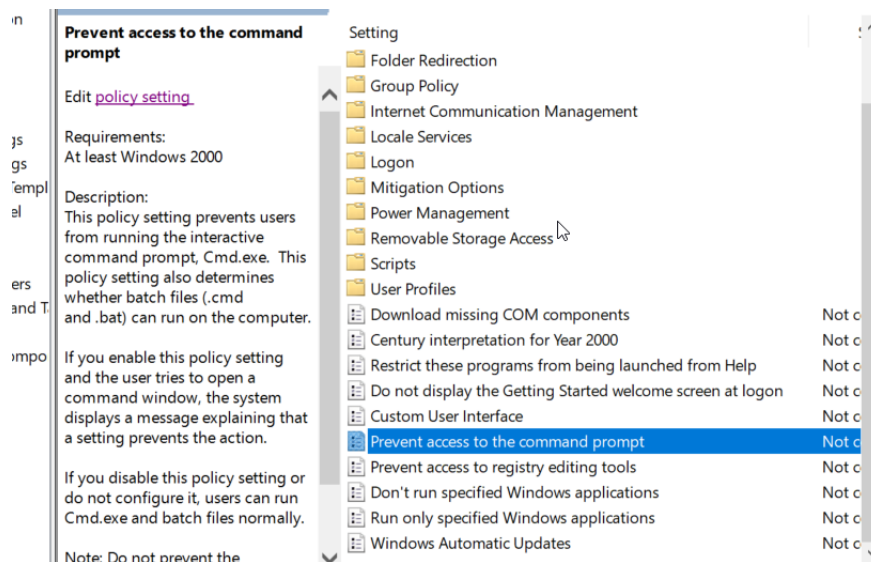


Step 3. Configure Command Prompt Restrictions

Navigated to the **System** settings within the Administrative Templates section of the IT Security Policy GPO. This area contains policy settings used to control access to Windows system tools and features.

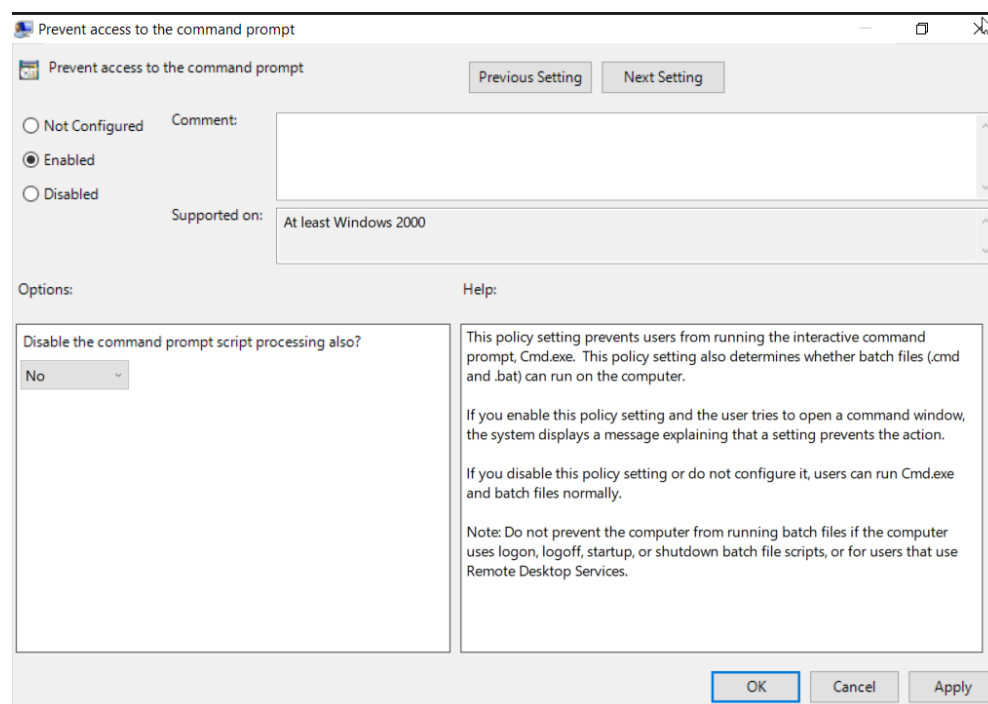


Located the **Prevent access to the command prompt** policy. This setting is used to restrict users from launching Command Prompt (CMD), helping reduce unauthorized command-line activity.



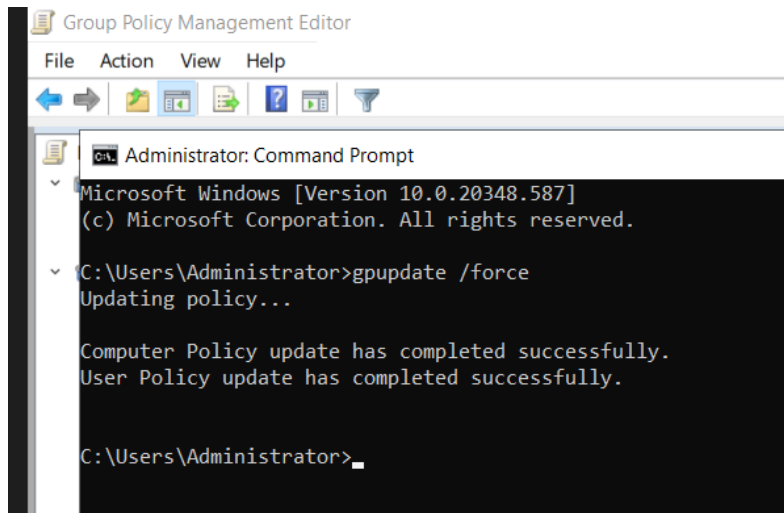
Note: Restricting access to Command Prompt helps prevent users from executing unauthorized commands and is commonly used in managed enterprise environments.

Enabled the **Prevent access to the command prompt** policy. Once applied, users within the IT OU will be unable to launch CMD.exe, reducing the risk of unauthorized system modifications.

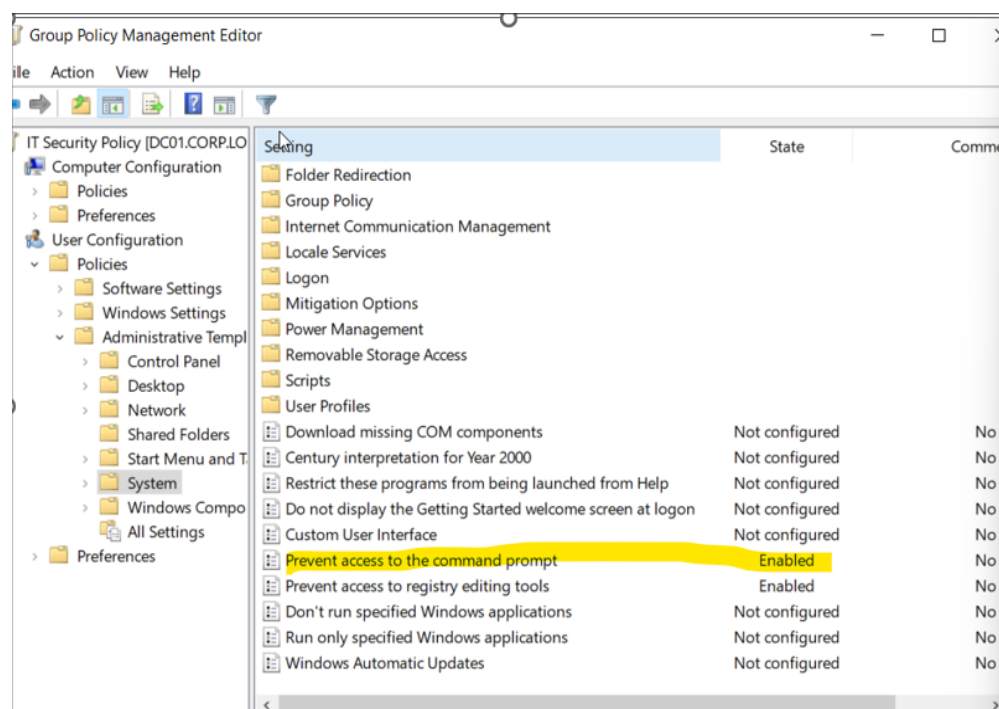


Step 4. Apply the Policy

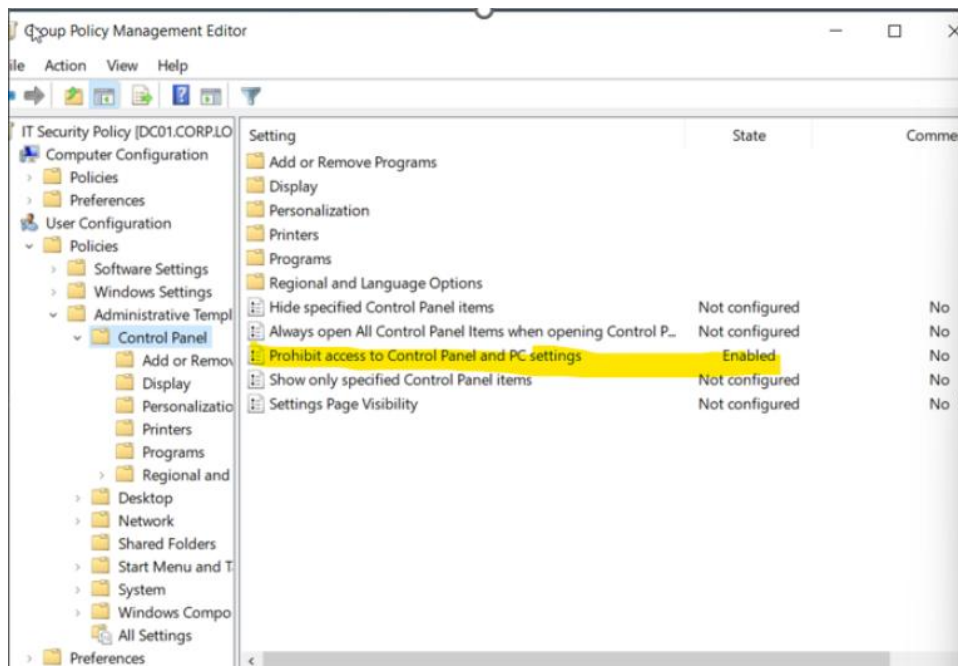
Executed the **gpupdate /force** command to immediately apply the newly configured Group Policy settings. The command completed successfully and updated both computer and user policies.



Verified that the **Prohibit access to Control Panel and PC settings** policy was successfully enabled within the IT Security Policy GPO.



Verified that the **Prevent access to the command prompt** policy was successfully enabled and configured within the IT Security Policy GPO.



Phase Summary

In this phase, I created a custom GPO and linked it to the IT OU. I configured security restrictions for Control Panel and Command Prompt, then applied and verified the policies through Group Policy Management. This demonstrated how administrators can manage user settings centrally in an Active Directory environment.

Phase 8: Password Reset and Account Unlock

Objective

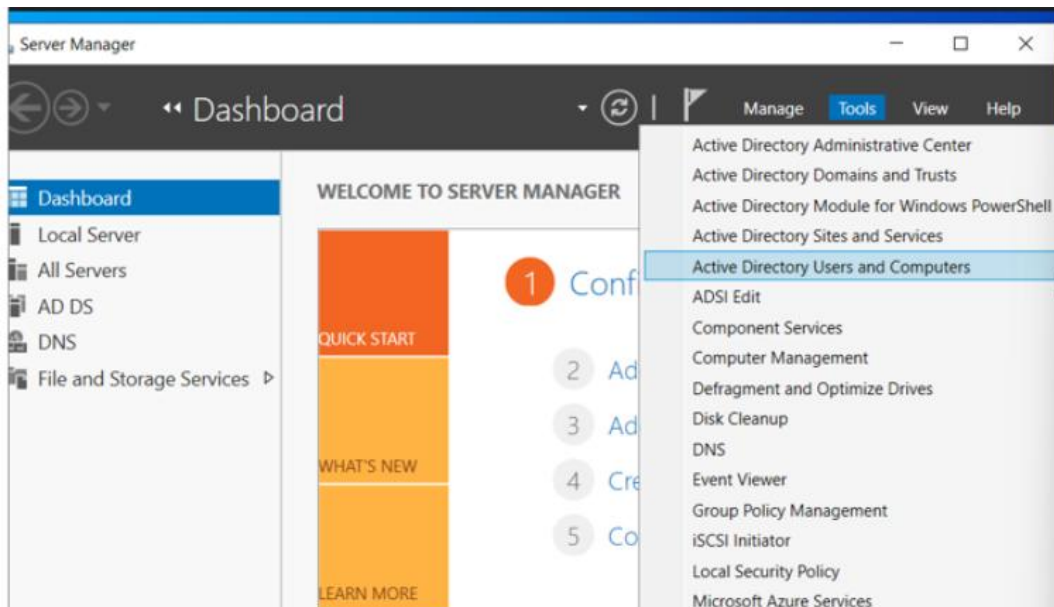
Simulate a common IT Help Desk scenario where a user forgets their password or becomes locked out after multiple failed login attempts.

Scenario

An employee from the IT department contacted the Help Desk because they forgot their password and could not access their domain account. The administrator was required to reset the password and restore account access.

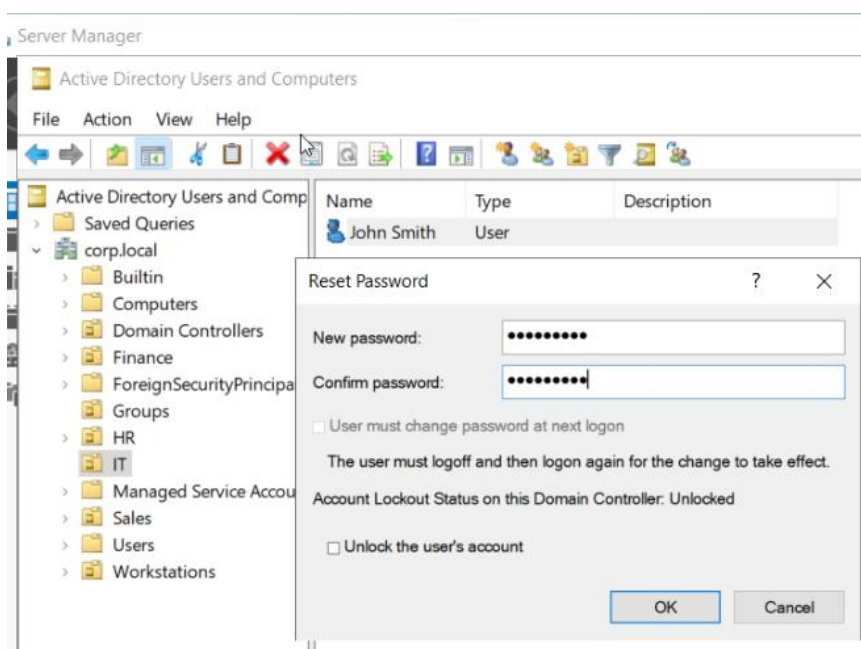
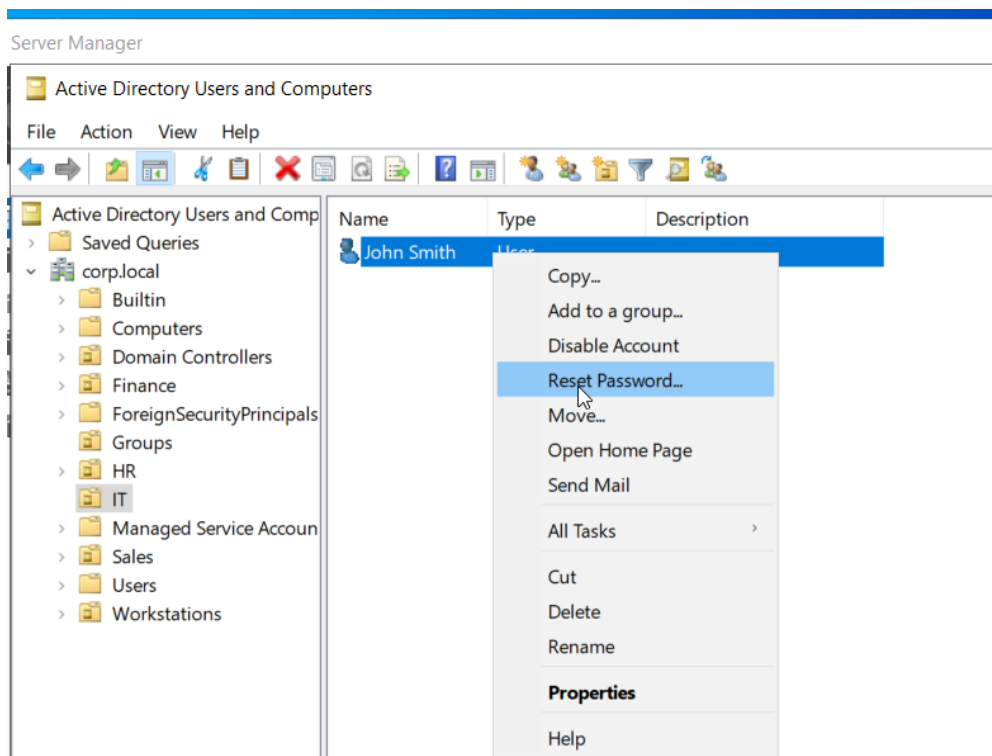
Step 1 – Open Active Directory Users and Computers (ADUC)

In this step, I opened Active Directory Users and Computers (ADUC), which is the primary administration tool used to manage users, groups, and organizational units within the domain.



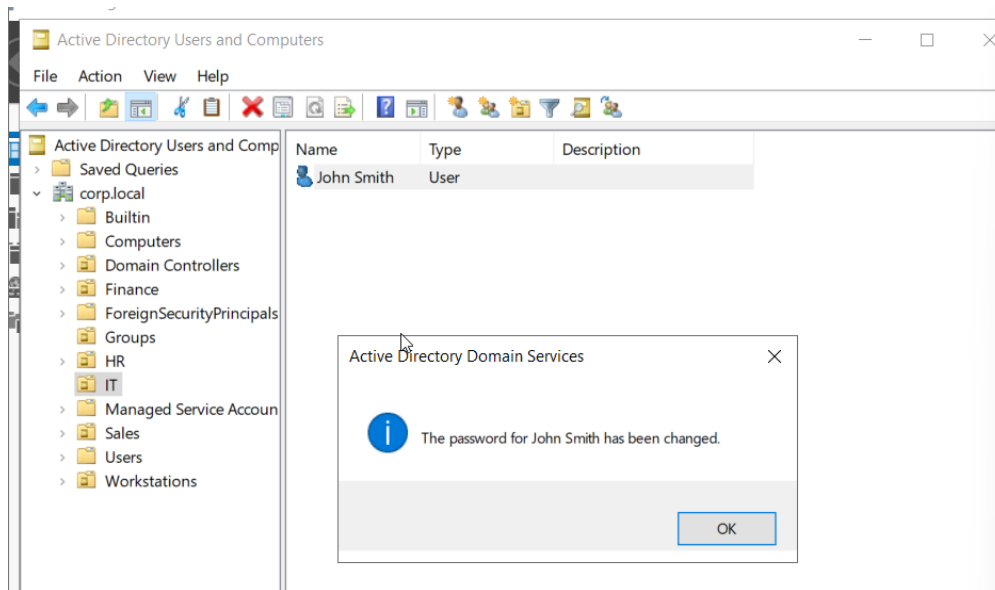
Step 2 -Reset the User Password

In this step, I performed a password reset for the user account. This is one of the most common tasks performed by Help Desk and System Administrators when users forget their passwords or are unable to log in.



Step 3 – Verify Account Status

Opened the user account properties and reviewed the account settings to confirm that the account remained active and accessible.



After resetting the password, I verified the account configuration to ensure that the user account remained enabled and that no additional restrictions were preventing access to the domain.

Phase Summary

In this phase, I successfully reset a domain user's password and verified that the account remained active and unlocked. This task reflects a typical Help Desk support procedure used to restore user access and maintain business continuity.

What I Learned

Through this project, I gained practical experience with Active Directory administration in Windows Server 2022. I learned how to deploy AD DS, create Organizational Units (OUs), manage users and groups, implement Role-Based Access Control (RBAC), configure shared folder permissions, and apply Group Policy Objects (GPOs).

I also practiced common IT Help Desk tasks such as password resets, account unlocks, user account management, and troubleshooting access issues within a domain environment.

Overall, this project provided hands-on experience with the core responsibilities typically performed by IT Support, Service Desk, and Junior System Administrators.